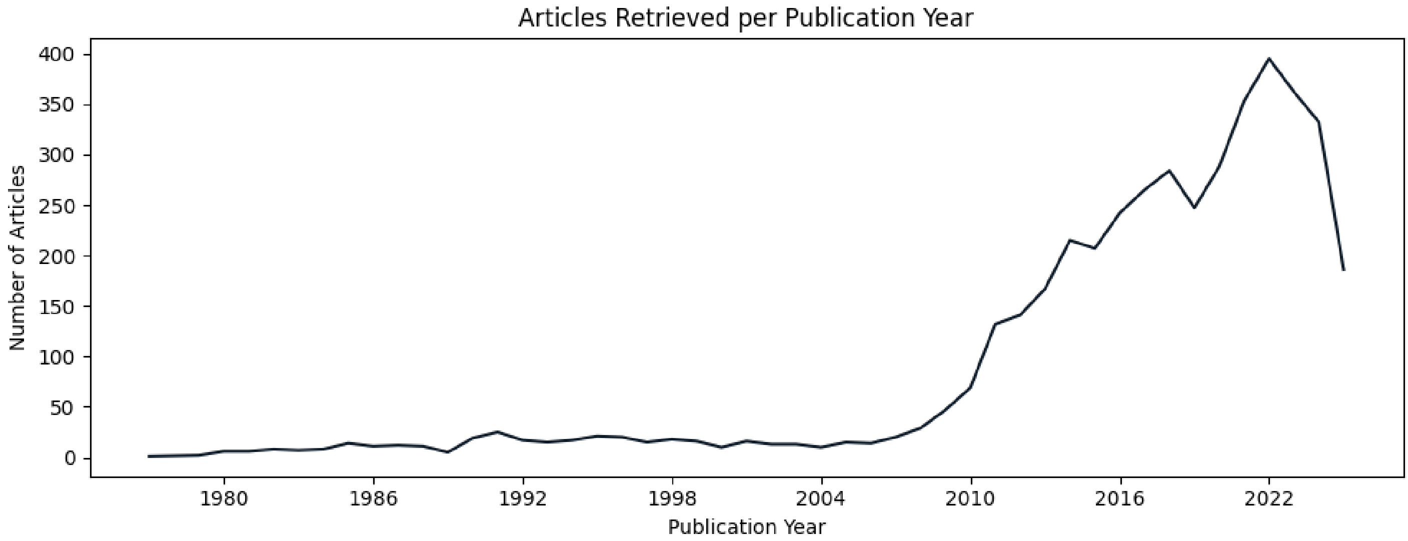
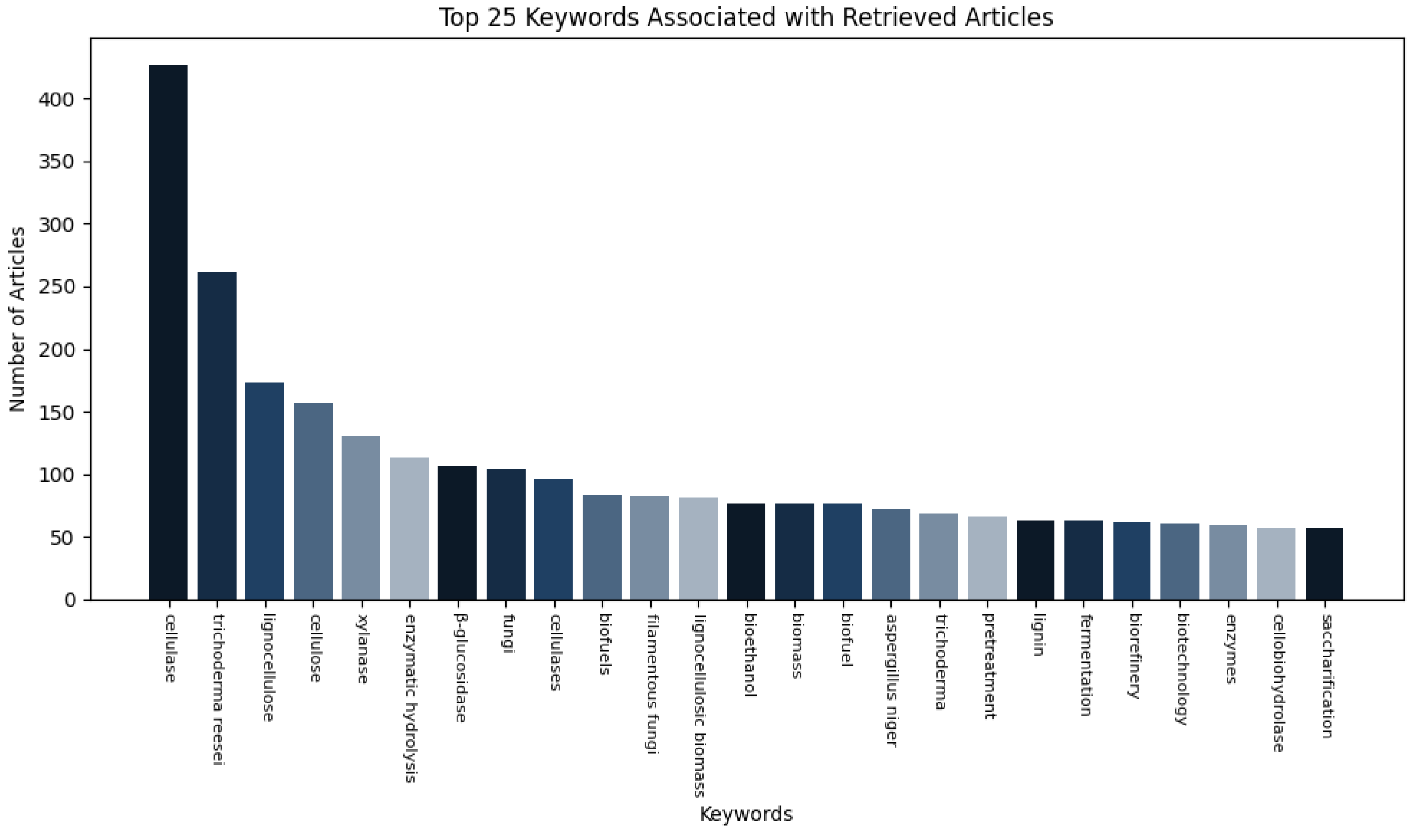
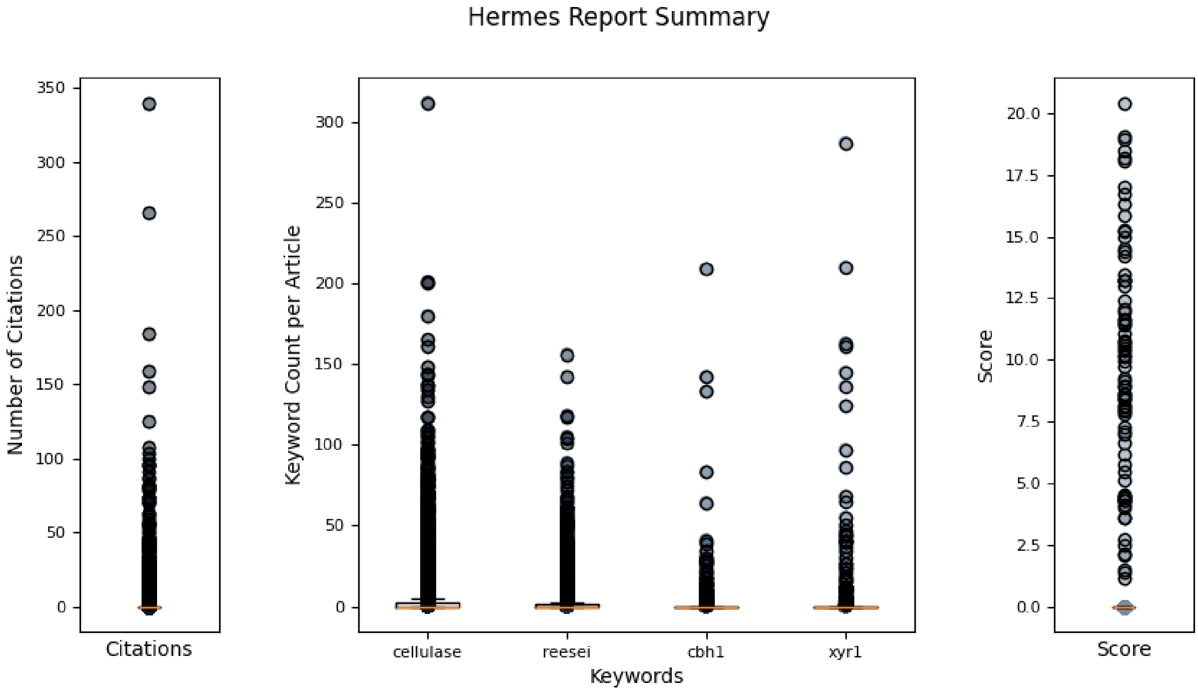
Literature Mining Report

Query Statistics

Query date: 2025-08-25
Query terms: "cellulase production trichoderma reesei"
Articles parsed/hits: 4354/4354 (100.00%)
Scoring keywords: cellulase, reesei, cbh1, xyr1 (Strict Mode)







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
32877410	2020	PLoS Genetics	Zheng, Fanglin et al.	trichoderma reesei xyr1 activates cellulase g...	19	20.44
35317826	2022	Microbial Cell	Mattam, Anu Jos et al.	factors regulating cellulolytic gene expressi...	0	19.05
38548869	2024	Communications	Zhu, Zhihua et al.	the protein methyltransferase trsam inhibits ...	4	18.94
36528622	2022	Biotechnology f	Shen, Linjing et al.	tailoring the expression of xyr1 leads to eff...	0	18.50
32765463	2020	Frontiers in Mi	Jiang, Xianzhan et al.	improved production of majority cellulases in...	0	18.18
34983541	2022	Microbial Cell	Wang, Yifan et al.	development of a powerful synthetic hybrid pr...	6	18.09
33606681	2021	PLoS Genetics	Wang, Lei et al.	interdependent recruitment of cyc8/tup1 and t...	10	17.05
36376415	2022	Scientific Repo	Arai, Toshiharu et al.	inducer-free cellulase production system base...	0	16.74
36732590	2023	Scientific Repo	Schalamun, Miri et al.	mapkinases regulate secondary metabolism, sex...	11	16.31
35852347	2022	Microbiology Sp	Wang, Zhixing et al.	functional characterization of sugar transpor...	7	15.83
24294104	2013	Current Genomic	Amore, Antonell et al.	regulation of cellulase and hemicellulase gen...	0	15.27
23152805	2012	PLoS ONE	Zhang, Jiwei et al.	ras gtpases modulate morphogenesis, sporulati...	0	15.19
25302561	2014	Molecular Micro	Lichius, Alexan et al.	nucleo-cytoplasmic shuttling dynamics of the ...	0	15.00
28497120	2017	mSphere	Stappler, Eva et al.	analysis of light- and carbon-specific transc...	28	14.50
30902998	2019	Scientific Repo	Rantasalo, Anss et al.	novel genetic tools that enable highly pure p...	27	14.38
32719779	2020	Frontiers in Bi	Zhang, Fei et al.	global reprogramming of gene transcription in...	0	14.21
31077201	2019	Microbial Cell	Liu, Pei et al.	enhancement of cellulase production in tricho...	16	13.47
25909478	2015	BMC Genomics	Lichius, Alexan et al.	genome sequencing of the trichoderma reesei q...	0	13.24
25626518	2015	Molecular Micro	Ghassemi, Sara et al.	the -importin kap8 (pse1/kap121) is required...	0	13.23
37488642	2023	Biotechnology f	Yan, Su et al.	role of cellulose response transporter-like p...	0	13.20
30406095	2018	Frontiers in Bi	Borin, Gustavo et al.	gene co-expression network reveals potential ...	0	13.00
38647894	2022	Bioresources an	Li, Ni et al.	induction of cellulase production by sr2+ in ...	0	12.41
38671155	2024	Scientific Repo	Schalamun, Miri et al.	the transcription factor ste12 influences gro...	2	12.08
32671045	2020	Frontiers in Bi	Meng, Qing-Shan et al.	engineering the effector domain of the artifi...	0	11.93
36977777	2023	Scientific Repo	Christopher, Me et al.	early cellular events and potential regulator...	0	11.63

Query Result 1/25

Trichoderma reesei XYR1 activates cellulase gene expression via interaction with the Mediator subunit TrGAL11 to recruit RNA polymerase II

Zheng, Fanglin; Cao, Yanli; Yang, Renfei; Wang, Lei; Lv, Xinxing; Zhang, Weixin; Meng, Xiangfeng; Liu, Weifeng; Butler, Geraldine

PLoS Genetics

Year: 2020 PMID: 32877410 doi: 10.1371/journal.pgen.1008979 Citations: 19 Score: 20.44

Keywords hits: cellulase: 67, reesei: 30, cbh1: 21, xyr1: 97

Abstract:
The ascomycete Trichoderma reesei is a highly prolific cellulase producer. While XYR1 (Xylanase regulator 1) has been firmly established to be the master activator of cellulase gene expression in T. reesei, its precise transcriptional activation mechanism remains poorly understood. In the present study, TrGAL11, a component of the Mediator tail module, was identified as a putative interacting partner of XYR1. Deletion of Trgal11 markedly impaired the induced expression of most (hemi)cellulase genes, but not that of the major -glucosidase encoding genes. This differential involvement of TrGAL11 in the full induction of cellulase genes was reflected by the RNA polymerase II (Pol II) recruitment on their core promoters, indicating that TrGAL11 was required for the efficient transcriptional initiation of the majority of cellulase genes. In addition, we found that TrGAL11 recruitment to cellulase gene promoters largely occurred in an XYR1-dependent manner. Although xyr1 expression was significantly tuned down without TrGAL11, the binding of XYR1 to cellulase gene promoters did not entail TrGAL11. These results indicate that TrGAL11 represents a direct in vivo target of XYR1 and may play a critical role in contributing to Mediator and the following RNA Pol II recruitment to ensure the induced cellulase gene expression.

Summary:
Altogether, the data indicate that Trgal11 plays an important role in mediating the induced expression of cellobiohydrolase and endoglucanase but not -glucosidase genes in T. reesei. To evaluate the relevant contribution of other putative tail module subunits to the induced cellulase gene expression, the identified Trmed3, Trmed5, and Trmed16 that have been shown to be nonessential in yeast , were individually deleted in T. reesei. ChIP-qPCR analyses revealed significantly higher XYR1 occupancy on all the relevant cellulase gene promoters including -glucosidase gene promoters in the Trgal11 deletion strain than that in QM9414. In addition, TrGAL11 and thus RNA Pol II binding to cellulase gene promoters specifically relied on the expression of XYR1 whereas the steady state XYR1 binding to its target regulatory sequences was subject to a regulation imposed by TrGAL11 recruitment. Regulation of eukaryotic Pol II transcription is carried out in many ways, from the DNA sequence and chromatin architecture to recruitment and regulation of large protein assemblies at the promoter . One possible explanation for the differential involvement of TrGAL11 and TrMED5 in the cellulase gene expression is that an as yet to be identified transcriptional activator synergizes with XYR1 in achieving core Mediator recruitment at -glucosidase gene promoters by engaging interactions with subunits other than TrGAL11 and TrMED5. An interesting finding in our present research was that XYR1 binding to cellulase gene promoters seems to be significantly enhanced although xyr1 expression itself was reduced in the absence of TrGAL11. All T. reesei strains used in this research were listed in. To assay vegetative growth, strains were precultured on minimal media agar plate for two days and then a slice of agar with the same area of growing mycelia of the corresponding strain was taken from the plate and inoculated on minimal media agar plates containing different carbon sources at 30C for 3 days or on malt extract agar plates incubated for 5 days. To determine T. reesei biomass accumulation in liquid MA medium with 1% glucose or Avicel as the sole carbon source, equal amounts of pre-cultured mycelia as determined by wet weight were inoculated into the indicated medium.

Figures:

Figure 1: TrGAL11 interacts with XYR1in vitro.

Figure 2: Trgal11disruption had little effect onT.reeseigrowth on plates or in liquid medium but reduced its conidia and pigment formation.

Figure 3: Deletion ofTrgal11compromised the fully induced production of cellobiohydrolase and endoglucanase but not of -glucosidase.

Figure 4: Deletion ofTrgal11resulted in a significant decrease in the transcription of cellobiohydrolase and endoglucanase but not -glucosidase genes.

Figure 5: Overexpression ofxyr1failed to restore the fully induced expression of cellulase genes in theTrgal11deletion mutant.

Figure 6: TrGAL11 was recruited to cellulase gene promoters in an XYR1-dependent manner.

Figure 7: Trgal11deletion resulted in a significantly higher XYR1 occupancy on cellulase gene promoters.

Figure 8: TrGAL11 plays a critical role in RNA Pol II recruitment on the core-promoter ofcbh/eggenes but not ofbglgenes.

Figure 9: A schematic model of how Mediator is recruited by XYR1 to participate in activating cellulase gene expression inT.reesei.

Mentioned biomedical entities:

Genes: CKM, FPA, GAL4, GCN4, HSF1, MED14, MED16, MED2, MED5, PIC, Rpb1, TATA, TFIIH, UPS, XYR1, actin, bgl1, bgl2, cbh1, cellulase, eg1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: E. coli, human

Tissues: Tail

Pathways: N/A

Query Result 2/25

Factors regulating cellulolytic gene expression in filamentous fungi: an overview

Mattam, Anu Jose; Chaudhari, Yogesh Babasaheb; Velankar, Harshad Ravindra

Microbial Cell Factories

Year: 2022 PMID: 35317826 doi: 10.1186/s12934-022-01764-x Citations: 0 Score: 19.05

Keywords hits: cellulase: 200, reesei: 105, cbh1: 17, xyr1: 68

Abstract:

The growing demand for biofuels such as bioethanol has led to the need for identifying alternative feedstock instead of conventional substrates like molasses, etc. Lignocellulosic biomass is a relatively inexpensive feedstock that is available in abundance, however, its conversion to bioethanol involves a multistep process with different unit operations such as size reduction, pretreatment, saccharification, fermentation, distillation, etc. The saccharification or enzymatic hydrolysis of cellulose to glucose involves a complex family of enzymes called cellulases that are usually fungal in origin. Cellulose hydrolysis requires the synergistic action of several classes of enzymes, and achieving the optimum secretion of these simultaneously remains a challenge. The expression of fungal cellulases is controlled by an intricate network of transcription factors and sugar transporters. Several genetic engineering efforts have been undertaken to modulate the expression of cellulolytic genes, as well as their regulators. This review, therefore, focuses on the molecular mechanism of action of these transcription factors and their effect on the expression of cellulases and hemicellulases.

Summary:

In fact, the overexpression of Ace4 led to a 22% increase in the expression of the major cellulase genes, and this effect was mediated in an Ace3-dependent manner . The deletion of Ace3 resulted in negligible expression of cbh1, cbh2, egl1, bgl1, and xyn3 in the recombinant strain as compared to the parental strain . Another approach for relieving catabolite repression involves the deletion of the alpha-tubulin tubB gene in T. reesei, which led to the upregulation of several cellulase and hemicellulase genes, as well as genes encoding transporters of cellobiose and other sugars, when grown in a medium containing either glucose or cellobiose, suggesting that tubB could be involved in sugar sensing as well, and therefore analogous to Cre1 . The repression of cellulase expression in presence of glucose in Aspergillus sp. It has been observed that Xpp1 regulates the transcription of hemicellulase genes only in the later stages of growth and no significant difference in xylanase activities was observed in an xpp1-disrupted strain or its parent strain, in the initial 72 h, reconfirming its role as a repressor of secondary metabolism . The SxlR transcription factor was identified due to its role in inhibiting the expression of the major xylanases such as xyn1, xyn2, xyn5, etc. As expected, the overexpression of the Ctf1 gene under the constitutive pdc1 promoter led to significant repression of cellulase genes . In another study, Rce2, a protein recently identified by a pull-down and mass spectrometry analysis was found to have similar binding sites in the promoter regions of target genes as Ace3, and its overexpression led to reduced cellulase expression i. e. Rce2 acted antagonistic to Ace3 in T. reesei with regards to cellulase induction and led to repressed cellulase and hemicellulase expression . The calcineurin-responsive zinc finger transcription factor 1 or Crz1 was identified in T. reesei by gene disruption . Despite significant progress made in studies related to the production of cellulases, the influence of sugar transporters on the degradation of cellulose or lignocellulosic biomass has not been investigated in detail. The deletion of the MFS protein Stp1 that transports cellobiose in T. reesei, resulted in the repression of both cellulase and hemicellulase genes, in presence of Avicel however, no effect on cellulase expression was observed when the strain was grown in cellobiose.

Figures:

Figure 1: Transcriptional regulators and transporters involved in cellulase expression in T. reesei and Penicillium sp

Figure 2: Comparative analysis of genome with respect to CAZymes in Trichoderma reesei , Penicillium funiculosum and Penicillium oxalicum 1142.

Mentioned biomedical entities:

Genes: Ace1, Ace2, Ace3, AceI, Are1, AreA, Azf1, B, BglR, CYC8, Cdt1, CdtC, Clr1, Clr2, Cre1, CreA, CreB, CreC, Crt1, Crz1, Ctf1, GH, GHs, Gal11, HapB, LaeA, MFS, Pac1, PacC, PalA, Stp1, TUP1, TacA, Vel2, Vel3, VelB, Vib1, XlnR, Xyr1, amdS, bgl1, bgl2, cbh1, cbh2, cellulase, eg1, pac1, pal, tubB, xlnR, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Chromolaena odorata, S. cerevisiae

Tissues: N/A

Pathways: N/A

Query Result 3/25

The protein methyltransferase TrSAM inhibits cellulase gene expression by interacting with the negative regulator ACE1 in Trichoderma reesei

Zhu, Zhihua; Zou, Gen; Chai, Shunxing; Xiao, Meili; Wang, Yinmei; Wang, Pingping; Zhou, Zhihua

Communications Biology

Year: 2024 PMID: 38548869 doi: 10.1038/s42003-024-06072-1 Citations: 4 Score: 18.94

Keywords hits: cellulase: 58, reesei: 48, cbh1: 34, xyr1: 34

Abstract:
Protein methylation is a commonly posttranslational modification of transcriptional regulators to fine-tune protein function, however, whether this regulation strategy participates in the regulation of lignocellulase synthesis and secretion in Trichoderma reesei remains unexplored. Here, a putative protein methyltransferase (TrSAM) is screened from a T. reesei mutant with the ability to express heterologous -glucosidase efficiently even under glucose repression. The deletion of its encoding gene trsam causes a significant increase of cellulase activities in all tested T. reesei strains, including transformants of expressing heterologous genes using cbh1 promotor. Further investigation confirms that TrSAM interacts with the cellulase negative regulator ACE1 via its amino acid residue Arg383, which causes a decrease in the ACE1-DNA binding affinity. The enzyme activity of a T. reesei strain harboring ACE1R383Q increases by 85.8%, whereas that of the strains with trsam or ace1 deletion increases by more than 100%. By contrast, the strain with ACE1R383K shows no difference to the parent strain. Taken together, our results demonstrate that TrSAM plays an important role in regulating the expression of cellulase and heterologous proteins initiated by cbh1 promotor through interacting with ACE1R383. Elimination and mutation of TrSAM and its downstream ACE1 alleviate the carbon catabolite repression (CCR) in expressing cellulase and heterologous protein in varying degrees. This provides a new solution for the exquisite modification of T. reesei chassis.

Summary:
In repressing medium, the pNPGase activities of atbg-A1, atbg-D3 and atbg-U10 were 5.36 0.69, 121.32 6.87, and 231.32 21.85 IU/mL, respectively, and the pNPGase activity of Rut-C30 was barely detectable.), which demonstrated that BglS could be expressed when glucose was used as the carbon source in atbg-D3 and atbg-U10 strains. To investigate the function of TrSAM in the regulation of cellulase production, interactions between TrSAM and widely known cellulase-related regulators were analyzed using the yeast two-hybrid system. The interaction between TrSAM-AD and ACE1-BD promoted yeast growth in the absence of histidine and adenine and resulted in X--gal activity, whereas the interaction between TrSAM-AD and Lae1-BD or other fused regulators did not. It has been testified that ACE1 and XYR1 may bind to the promoters of xyr1, cbh1 and other cellulase or hemicellulase genes competitively. In this study, we found that the methyltransferase TrSAM participates in the repression of cellulase synthesis by the interaction with the negative regulator ACE1 and thus strengthening its binding to the promotor regions of target genes. Although it is possible that the low expression level of ACE1 under inducing condition has weakened its interaction with TrSAM, which further impairs its repressing effect on cellulase expression, the mutation of 383 residue arginine to glutamine or deletion of trsam has been demonstrated to improve cellulase production regardless of inducing or repressing conditions (Fig.

Figures:

Figure 1: Evaluation of enzymatic activity and protein production by transformants expressing BglS in inducing and repressing conditions.

Figure 2: Schematic diagram of large missing fragments in atbg-D3 and atbg-U10.

Figure 3: Interaction analysis of protein methyltransferaseTrSAM and ACE1.

Figure 4: Evaluation of cellulase production of Rut-C30 and its mutants.

Figure 5: Comparison of ACE1 and ACE1R383Qabilities to bind DNA.

Figure 6: Schematic model ofTrSAM-mediated regulatory mechanism during carbon catabolite repression inT. reesei.

Mentioned biomedical entities:

Genes: ACE1, ACE2, BglR, BglS, CBH1, CCR, Cre1, FPA, GAL4, GATA4, GST, Hda1, Lae1, Mig1, MyoD, PRC2, RK, Roc1, Vel1, XYR1, ace1, bglR, cbh1, cellulase, cre1, histone, p53

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: E. coli

Tissues: N/A

Pathways: N/A

Query Result 4/25

Tailoring the expression of Xyr1 leads to efficient production of lignocellulolytic enzymes in Trichoderma reesei for improved saccharification of corncob residues

Shen, Linjing; Yan, Aiqin; Wang, Yifan; Wang, Yubo; Liu, Hong; Zhong, Yaohua

Biotechnology for Biofuels and Bioproducts

Year: 2022 PMID: 36528622 doi: 10.1186/s13068-022-02240-9 Citations: 0 Score: 18.50

Keywords hits: cellulase: 96, reesei: 50, cbh1: 28, xyr1: 124

Abstract:

The filamentous fungus Trichoderma reesei is extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulase gene expression. However, the role of Xyr1 in the regulation of cellulase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolytic enzyme production.

Summary:

Furthermore, the overexpression of xyr1 with Pegl2 also remarkably improved the levels of accessory proteins involved in lignocellulose degradation and thus promoted the saccharification efficiency toward acid-pretreated corncob residues. To systematically investigate the effect of differential expression of xyr1 on cellulase production in T. reesei, the xyr1 overexpression cassettes driven by the cellulose-inducible promoters Pegl2, Pcbh1, and the strong constitutive promoter Pcdna1 were constructed and precisely integrated by homologous recombination into the hph locus of T. reesei QEB4, respectively. It was found that xyr1 was significantly overexpressed in the strains, with the maximum transcript levels in the QCDX strain, followed by QCBX and QE2X, which were 47.2-, 12.6-, and 5.8-fold higher than the parental strain QEB4, respectively. Next, the three xyr1 overexpression strains QE2X, QCBX, and QCDX were cultured on minimal medium plates containing glucose, glycerol, and lactose as the carbon source, respectively, as well as on PDA plates to investigate the effect of differential xyr1 overexpression on mycelial phenotypes of T. reesei. Taken together, these results demonstrate that cellulase gene transcription is precisely regulated by the xyr1 transcript abundance, i. e., the moderately strong overexpression of xyr1 with the Pegl2 promoter, but not the stronger overexpression with the Pcbh1 or Pcdna1 promoter, could lead to more pronounced improvement in cellulase expression. Furthermore, transcript analysis of several transcription factors involved in cellulase expression were performed. The total cellulase activity of QE2X reached 8.4 IU/mL, 3.2-fold higher than that of the parental strain, using filter paper activity as a characterization method, while the FPA in QCBX and QCDX was only elevated 0.9- and 0.8-fold compared to that of the parental strain, respectively. These results indicate that the Pegl2-driven overexpression of xyr1 can optimize the lignocellulolytic enzyme system of T. reesei by improving the expression of those accessory proteins, which is beneficial for efficient degradation of lignocellulosic biomass. In T. reesei, genetic engineering of transcriptional regulators has been used as a powerful tool for enhancing cellulase production.

Figures:

Figure 1: Construction and validation of the xyr1 overexpression strains.

Figure 2: Detection of the cellulase secretion by T. reesei QEB4 and the xyr1 overexpression strains in different culture plates.

Figure 3: Transcript analysis by RT-qPCR for genes encoding cellulases and regulators.

Figure 4: Transcript and CHART analysis of T. reesei QEB4 and the xyr1 overexpression strains.

Figure 5: Transcript analysis of the genes encoding the ER-associated components by RT-qPCR.

Figure 6: Cellulase activities and total extracellular proteins of T. reesei QEB4 and the xyr1 overexpression strains.

Figure 7: Saccharification of different pretreated corncob residues by T. reesei QEB4 and QE2X.

Figure 8: Transcript analysis for genes encoding accessory proteins of T. reesei QEB4 and QE2X by RT-qPCR.

Figure 9: Schematic summary showing the proposed model for the dosage regulation of xyr1 in T. reesei .

Mentioned biomedical entities:

Genes: Ace1, Ace2, Ace3, BGL1, BglR, Cip1, Cip2, Cre1, Crz1, EG2, ER, FPA, Pac1, Ppdc, Rce1, Sec16, Swo1, Vib1, Xyr1, ace1, actin, bgl1, cbh1, cbh2, cellulase, cip1, cip2, cre1, histone, hrd1, pdi1, ptrA, swo1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 5/25

Improved Production of Majority Cellulases in Trichoderma reesei by Integration of cbh1 Gene From Chaetomium thermophilum

Jiang, Xianzhang; Du, Jiawen; He, Ruonan; Zhang, Zhengying; Qi, Feng; Huang, Jianzhong; Qin, Lina

Frontiers in Microbiology

Year: 2020 PMID: 32765463 doi: 10.3389/fmicb.2020.01633 Citations: 0 Score: 18.18

Keywords hits: cellulase: 70, reesei: 117, cbh1: 209, xyr1: 10

Abstract:

Lignocellulose is an abundant waste resource and has been considered as a promising material for production of biofuels or other valuable bio-products. Currently, one of the major bottlenecks in the economic utilization of lignocellulosic materials is the cost-efficiency of converting lignocellulose into soluble sugars for fermentation. One way to address this problem is to seek superior lignocellulose degradation enzymes or further improve current production yields of lignocellulases. In the present study, the lignocellulose degradation capacity of a thermophilic fungus Chaetomium thermophilum was firstly evaluated and compared to that of the biotechnological workhorse Trichoderma reesei. The data demonstrated that compared to T. reesei, C. thermophilum displayed substantially higher cellulose-utilizing efficiency with relatively lower production of cellulases, indicating that better cellulases might exist in C. thermophilum. Comparison of the protein secretome between C. thermophilum and T. reesei showed that the secreted protein categories were quite different in these two species. In addition, to prove that cellulases in C. thermophilum had better enzymatic properties, the major cellulase cellobiohydrolase I (CBH1) from C. thermophilum and T. reesei were firstly characterized, respectively. The data showed that the specific activity of C. thermophilum CBH1 was about 4.5-fold higher than T. reesei CBH1 in a wide range of temperatures and pH. To explore whether increasing CBH1 activity in T. reesei could contribute to improving the overall cellulose-utilizing efficiency of T. reesei, T. reesei cbh1 gene was replaced with C. thermophilum cbh1 gene by integration of C. thermophilum cbh1 gene into T. reesei cbh1 gene locus. The data surprisingly showed that this gene replacement not only increased the cellobiohydrolase activities by around 4.1-fold, but also resulted in stronger induction of other cellulases genes, which caused the filter paper activities, Azo-CMC activities and -glucosidase activities increased by about 2.2, 1.9, and 2.3-fold, respectively. The study here not only provided new resources of superior cellulases genes and new strategy to improve the cellulase production in T. reesei, but also contribute to opening the path for fundamental research on C. thermophilum.

Summary:

One way to overcome these drawbacks is the introduction of lignocellulases encoding genes from thermophilic fungi into mesophilic species, in which a variety of genetic tools have been developed, to combine the advantages of mesophilic and thermophilic properties. The soft-rot fungus Trichoderma reesei is a model organism for plant biomass degradation and widely used in industry for the production of cellulases and xylanases due to the large capacity of hydrolytic enzymes secretion. Further analysis of the extracellular protein levels in the culture supernatants of C. thermophilum and T. reesei showed that under Avicel, the levels of secreted proteins in C. thermophilum were slightly higher than that in T. reesei in the first 3 days. To investigate whether increased cellobiohydrolase activities in T. reesei could improve its cellulose utilization, recombinant T. reesei strain in which the native cbh1 gene was replaced with cbh1 gene from C. thermophilum was created and named as Tr-Ctcbh1. Three independent Tr-Ctcbh1 and Cpyr4 recombinant strains were selected for the measurement of the secreted protein levels, CBH activities, and the filter paper activities under Avicel. Considering that the certain oxidants could make lignocellulose more susceptible for enzymatic degradation, the existence of such amounts of oxidoreductases in the supernatants of C. thermophilum culture implied that in addition to contain higher efficient cellulases, C. thermophilum might display a different mode of action to degrade lignocelluloses. In T. reesei, it has been generally believed that -glucosidase is a barrier for its cellulose degradation capacity because the extracellular -glucosidase comprises only about 1% of the total T. reesei cellulases complex, based on which, many efforts have been conducted to increase enzyme efficiency of hydrolyzing cellulosic substrates by increasing -glucosidase amount in the cellulases complex from T. reesei. The results that the FPA activities, the Azo-CMC activities and -glucosidase activities in Tr-Ctcbh1 strains were significantly higher than those in Cpyr4 strains, and besides, the transcriptional levels of cbh2, eg1, eg2, and bgl1 genes were also higher in Tr-Ctcbh1 strains, which further proved that the induction and expression of most cellulases genes in T. reesei could be improved by only enhancing CBH1 activities. The increased expression levels of xyr1, cre1, and ace3 in Tr-Ctcbh1 strains suggested that an extra inducer might be generated in T. reesei when response to cellulose with more cellobiohydrolases in its cellulases system.

Figures:

Figure 1: Phenotypic analysis of C. thermophilum and T. reesei when response to cellulose.

Figure 2: The extracellular protein categories in the supernatants of cellulose cultures were different between C. thermophilum and T. reesei.

Figure 3: Expression of Tr-CBH1 and Ct-CBH1 in T. reesei using cellulase-free background system.

Figure 4: Ct-CBH1 showed higher specific activity compared to Tr-CBH1.

Figure 5: Ct-CBH1 showed higher optimum temperature and better temperature stability compared to Tr-CBH1.

Figure 6: Replacement of Tr- cbh1 with Ct- cbh1 gene in T. reesei increased the production of cellulases.

Figure 7: The activities of other cellulases were also increased in Tr-Ct cbh1 strains.

Figure 8: Increased CBH1 activities in T. reesei enhanced the induction and expression of the major cellulases genes.

Figure 9: The higher production of cellulases in Tr-Ct cbh1 strains was caused by increased expression level of xyr1.

Figure 10: Graphic summary of this study.

Mentioned biomedical entities:
Genes: CBH, CBH1, FPA, MUC, ace1, ace2, actin, bgl1, cbh1, cbh2, cellulase, cre1, eg1, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: C. thermophilum

Tissues: N/A

Pathways: N/A

Query Result 6/25

Development of a powerful synthetic hybrid promoter to improve the cellulase system of Trichoderma reesei for efficient saccharification of corncob residues

Wang, Yifan; Liu, Ruiyan; Liu, Hong; Li, Xihai; Shen, Linjing; Zhang, Weican; Song, Xin; Liu, Weifeng; Liu, Xiangmei; Zhong, Yaohua

Microbial Cell Factories

Year: 2022 PMID: 34983541 doi: 10.1186/s12934-021-01727-8 Citations: 6 Score: 18.09

Keywords hits: cellulase: 44, reesei: 61, cbh1: 133, xyr1: 5

Abstract:

The filamentous fungus Trichoderma reesei is a widely used workhorse for cellulase production in industry due to its prominent secretion capacity of extracellular cellulolytic enzymes. However, some key components are not always sufficient in this cellulase cocktail, making the conversion of cellulose-based biomass costly on the industrial scale. Development of strong and efficient promoters would enable cellulase cocktail to be optimized for bioconversion of biomass.

Summary:

Therefore, optimization of T. reesei cellulase system by overexpressing specific essential enzymes directed by powerful promoters is a promising solution to improve the lignocellulose hydrolysis efficiency. In this study, a novel synthetic hybrid promoter Pcc was constructed by fusing the modified activation region of the strongest inducible promoter Pcbh1 to the core region of a strong constitutive promoter Pcdna1 after identifying their function. Then, the promoters Pcbh1-dc and Pcbh1-cr were fused with egfp to construct the corresponding cassettes, respectively. Based on the observation that the total cellulase activity of T. reesei can be improved by overexpressing specific cellulases directed by Pcc, this hybrid promoter was demonstrated to be efficient in optimizing cellulase system. Given that the individual overexpression of BGLA and EG2 driven by Pcc could led to improvement of the activities of BGL and EG, respectively, and both of them resulted in the FPA augmentation, a hypothesis was proposed that construction of strains co-overexpressing BGLA and EG2 directed by Pcc could further optimize the T. reesei enzymatic system to improve the cellulolytic ability. Then, QPEB29 and QPEB70 were cultured in CPM for 7 days at 30 C for cellulase production. The selectable marker hygromycin B gene fragment was amplified by hph-F/hph-R. Then the five differently truncated promoters of cdna1 were respectively employed to fuse with the upstream homologous arm of cbh1, hph and cbh1-ORF through Double-joint PCR, and primer pair cbh1-UF/cbh1-R was used to amplify the final cassettes.

Figures:

Figure 1: Nucleotide sequences of the P cdna1 (a) and P cbh1 (b).

Figure 2: Identification of the truncated fragments of P cdna1 fused with cbh1 as the reporter gene.

Figure 3: Identification of P cbh1 activation region (AR) using egfp as the reporter gene.

Figure 4: Construction of hybrid promoter P cc .

Figure 5: Expression of bglA driven by P cc resulted in increase of BGL activity and FPA.

Figure 6: Expression of eg2 driven by P cc resulted in increase of EG activity and FPA.

Figure 7: Co-overexpression of bglA and eg2 driven by P cc in T. reesei resulted in increase of FPA.

Figure 8: Saccharification of different pretreated corncob residues using the extracellular enzymes from QP4 and the transformants co-overexpressing bglA and eg2 driven by P cc .

Mentioned biomedical entities:

Genes: AR, Ace1, Ace2, BGL1, BGLA, CBH, CBH1, Cre1, EG2, FPA, Hap5, PBC1, PBD1, PD1, TATA, Xyr1, bgl1, cbh1, cellulase, gpdA, hph, pdc, ptrA

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. niger, Penicillium decumbens

Tissues: N/A

Pathways: N/A

Query Result 7/25

Interdependent recruitment of CYC8/TUP1 and the transcriptional activator XYR1 at target promoters is required for induced cellulase gene expression in Trichoderma reesei

Wang, Lei; Zhang, Weixin; Cao, Yanli; Zheng, Fanglin; Zhao, Guolei; Lv, Xinxing; Meng, Xiangfeng; Liu, Weifeng; Butler, Geraldine

PLoS Genetics

Year: 2021 PMID: 33606681 doi: 10.1371/journal.pgen.1009351 Citations: 10 Score: 17.05

Keywords hits: cellulase: 74, reesei: 33, cbh1: 3, xyr1: 86

Abstract:

Cellulase production in filamentous fungus Trichoderma reesei is highly responsive to various environmental cues involving multiple positive and negative regulators. XYR1 (Xylanase regulator 1) has been identified as the key transcriptional activator of cellulase gene expression in T. reesei. However, the precise mechanism by which XYR1 achieves transcriptional activation of cellulase genes is still not fully understood. Here, we identified the TrCYC8/TUP1 complex as a novel coactivator for XYR1 in T. reesei. CYC8/TUP1 is the first identified transcriptional corepressor complex mediating repression of diverse genes in Saccharomyces cerevisiae. Knockdown of Trcyc8 or Trtup1 resulted in markedly impaired cellulase gene expression in T. reesei. We found that TrCYC8/TUP1 was recruited to cellulase gene promoters upon cellulose induction and this recruitment is dependent on XYR1. We further observed that repressed Trtup1 or Trcyc8 expression caused a strong defect in XYR1 occupancy and loss of histone H4 at cellulase gene promoters. The defects in XYR1 binding and transcriptional activation of target genes in Trtup1 or Trcyc8 repressed cells could not be overcome by XYR1 overexpression. Our results reveal a novel coactivator function for TrCYC8/TUP1 at the level of activator binding, and suggest a mechanism in which interdependent recruitment of XYR1 and TrCYC8/TUP1 to cellulase gene promoters represents an important regulatory circuit in ensuring the induced cellulase gene expression. These findings thus contribute to unveiling the intricate regulatory mechanism underlying XYR1-mediated cellulase gene activation and also provide an important clue that will help further improve cellulase production by T. reesei.

Summary:

Altogether, the data indicate that TrCYC8/TUP1 plays an important role in mediating the induced cellulase gene expression. To test whether TrCYC8/TUP1 is recruited to the cellulase gene promoter and directly involved in activating cellulase gene expression, chromatin immunoprecipitation followed by quantitative PCR on selected promoter regions was performed using TrTUP1 antibody to determine TrTUP1 occupancy on cellulase gene promoters. In all these data suggest that TrCYC8/TUP1 is involved in remodeling nucleosomes positioned in cellulase gene promoters to activate their transcription although more precise regulatory mechanism needs to be investigated. The results of this study show that a TrCYC8/TUP1 complex exists in T. reesei, and is required for high-level activation of cellulase genes in vivo. Nonetheless, the impaired cellulase gene expression with repressed Trcyc8 or Trtup1 in this study implicates that the complex may act in a different mode from that of CRE1 in controlling the cellulolytic response in T. reesei although possible interactions between CRE1 and TrCYC8/TUP1 can not be excluded. Interestingly, the CYC8/TUP1 complex has been also reported to function in gene activation. Our observations that TrCYC8/TUP1 is absolutely required for induced cellulase gene expression and is only recruited to cellulase promoters upon induction indicate that a different mode of activation by TrCYC8/TUP1 is adopted. Nevertheless, there has been no report on CYC8/TUP1 function in gene activation in filamentous fungi although RCM1/RCO1, the CYC8/TUP1 homolog in Neurospora crassa, has been reported to mediate specific repression of the frequency gene , . Considering the critical role of XYR1 in the induced cellulase gene expression, it is reasonable to attribute the observed defect in transcriptional activation of cellulase genes with repressed Trtup1 or Trcyc8 to the impaired XYR1 binding to its target genes.

Figures:

Figure 1: Evolutionarily conserved CYC8/TUP1 complex exists inT.reesei.

Figure 2: Knockdown ofTrcyc8andTrtup1by conditional repression.

Figure 3: Knockdown ofTrcyc8orTrtup1compromisedT.reeseigrowth and conidiation on agar plates but not in liquid media.

Figure 4: Knockdown ofTrcyc8andTrtup1abolished cellulase production.

Figure 5: Trcyc8orTrtup1repression resulted in a significant decrease in the transcription of cellulase gene andxyr1transcription.

Figure 6: TrTUP1 was recruited to cellulase gene promoters upon induction in an XYR1-dependent manner.

Figure 7: XYR1 overexpression failed to rescue the defective cellulase production in Ptcu1-Trcyc8KDor Ptcu1-Trtup1.

Figure 8: Efficient binding of XYR1 to cellulase genes in vivo relied on TrTUP1.

Figure 9: The effect ofTrcyc8orTrtup1repression on the association of histone H4 with cellulase gene promoters upon cellulose induction.

Figure 10: A working model for interdependent binding of TrCYC8/TUP1 and XYR1 on cellulase gene promoters to mediate their induced expression.

Mentioned biomedical entities:

Genes: CBH1, CEL7A, CRE1, CYC8, Cellulase, IP, LexA, MA, MIG1, PIC, SAGA, SspI, TUP1, WD40, XYR1, Xyr1, actin, arg1, cbh1, cellulase, cre1, cyc1, histone, tup1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Escherichia coli, S. cerevisiae

Tissues: N/A

Pathways: N/A

Query Result 8/25

Inducer-free cellulase production system based on the constitutive expression of mutated XYR1 and ACE3 in the industrial fungus Trichoderma, breesei

Arai, Toshiharu; Ichinose, Sakurako; Shibata, Nozomu; Kakeshita, Hiroshi; Kodama, Hiroshi; Igarashi, Kazuaki; Takimura, Yasushi

Scientific Reports

Year: 2022 PMID: 36376415 doi: 10.1038/s41598-022-23815-4 Citations: 0 Score: 16.74

Keywords hits: cellulase: 144, reesei: 25, cbh1: 10, xyr1: 136

Abstract:

Trichoderma, bc toggle=yesreesei is a widely used host for producing cellulase and hemicellulase cocktails for lignocellulosic biomass degradation. Here, we report a genetic modification strategy for industrial T., bc toggle=yesreesei that enables enzyme production using simple glucose without inducers, such as cellulose, lactose and sophorose. Previously, the mutated XYR1V821F or XYR1A824V was known to induce xylanase and cellulase using only glucose as a carbon source, but its enzyme composition was biased toward xylanases, and its performance was insufficient to degrade lignocellulose efficiently. Therefore, we examined combinations of mutated XYR1V821F and constitutively expressed CRT1, BGLR, VIB1, ACE2, or ACE3, known as cellulase regulators and essential factors for cellulase expression to the T., bc toggle=yesreesei E1AB1 strain that has been highly mutagenized for improving enzyme productivity and expressing a -glucosidase for high enzyme performance. The results showed that expression of ACE3 to the mutated XYR1V821F expressing strain promoted cellulase expression. Furthermore, co-expression of these two transcription factors also resulted in increased productivity, with enzyme productivity 1.5-fold higher than with the conventional single expression of mutated XYR1V821F. Additionally, that productivity was 5.5-fold higher compared to productivity with an enhanced single expression of ACE3. Moreover, although the DNA-binding domain of ACE3 had been considered essential for inducer-free cellulase production, we found that ACE3 with a partially truncated DNA-binding domain was more effective in cellulase production when co-expressed with a mutated XYR1V821F. This study demonstrates that co-expression of the two transcription factors, the mutated XYR1V821F or XYR1A824V and ACE3, resulted in optimized enzyme composition and increased productivity.

Summary:

Consequently, we found that the E1AB1-XA3 strain, which combines the constitutive expression of the mutated XYR1V821F and ACE3, makes it possible to produce xylanases and a high yield of cellulases, even under inducer-free conditions. The E1AB1-XA3 strain had high cellulase productivity even in the absence of inducers, indicating that a combination of ACE3 and mutated XYR1V821F is effective for an increase in cellulase activity. However, the productivity of the E1AB1-XA3fl strain with FL-ACE3 was 14% lower than that of the E1AB1-XA3 strain with PT-ACE3 (Fig. These results suggested that a certain level of total XYR1 or mutated XYR1, such as mutated XYR1V821F or XYR1A824V, was required for high cellulase production. These results also indicated that constitutive expression of the mutated XYR1 exhibiting a glucose-blind phenotype and ACE3 was necessary for high cellulase production in the absence of inducers. Additionally, the partial truncation of the DNA-binding domain of ACE3 and disruptions of ace1 and rce1 contributed to high protein productivity, and the genotype of the E1AB1-XA3 strain was found to be the most effective combination in this study. We analyzed the gene expression and activity of cellulase and xylanase in a single expression of XYR1V821F, a single expression of PT-ACE3, and co-expression of XYR1V821F and PT-ACE3. Owing to the effect of the XYR1V821F mutation, the transcription of major cellulases in the E1AB1-X strain was approximately four orders of magnitude higher than that of the parent strain E1AB1. Enhanced cellulase productivity using glucose as the sole carbon source was achieved by constitutive expression of mutated XYR1V821F or XYR1A824V and ACE3, especially ACE3, having a partially truncated DNA-binding domain. In this report, co-expression of XYR1V821F with cellulase regulation-related factors, CRT1, BGLR, VIB1, or ACE2, did not increase cellulase expression.

Figures:

Figure 1: Effects of constitutive expression of V821F mutated XYR1 and disruption of repressors on glucose culture.

Figure 2: Co-expression of factors involved in cellulase transcription in the E1AB1-X strain.

Figure 3: Confirmation of the genetic combinations required for high cellulase production under inducer-free conditions.

Figure 4: Analysis of gene expression, enzyme activity, composition, and saccharification of microcrystalline cellulose.

Mentioned biomedical entities:

Genes: ACE2, ACE3, ARE1, BXL1, CBH, CBH1, CBH2, CIP2, CRE1, CRT1, CRZ1, CTF1, Cellulase, EG1, Gal4, HD, KC, LAE1, MP, PAC1, RCE1, STR1, VEL1, VIB1, XYN1, XYN2, XYR1, ace1, ace2, act1, cbh1, cbh2, cellulase, cre1, pgk1, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: tube

Pathways: N/A

Query Result 9/25

MAPkinases regulate secondary metabolism, sexual development and light dependent cellulase regulation in Trichoderma reesei

Schalamun, Miriam; Beier, Sabrina; Hinterdobler, Wolfgang; Wanko, Nicole; Schinnerl, Johann; Brecker, Lothar; Engl, Dorothea Elisa; Schmoll, Monika

Scientific Reports

Year: 2023 PMID: 36732590 doi: 10.1038/s41598-023-28938-w Citations: 11 Score: 16.31

Keywords hits: cellulase: 48, reesei: 40, cbh1: 15, xyr1: 9

Abstract:

The filamentous fungus Trichoderma reesei is a prolific producer of plant cell wall degrading enzymes, which are regulated in response to diverse environmental signals for optimal adaptation, but also produces a wide array of secondary metabolites. Available carbon source and light are the strongest cues currently known to impact secreted enzyme levels and an interplay with regulation of secondary metabolism became increasingly obvious in recent years. While cellulase regulation is already known to be modulated by different mitogen activated protein kinase (MAPK) pathways, the relevance of the light signal, which is transmitted by this pathway in other fungi as well, is still unknown in T. reesei as are interconnections to secondary metabolism and chemical communication under mating conditions. Here we show that MAPkinases differentially influence cellulase regulation in light and darkness and that the Hog1 homologue TMK3, but not TMK1 or TMK2 are required for the chemotropic response to glucose in T. reesei. Additionally, MAPkinases regulate production of specific secondary metabolites including trichodimerol and bisorbibutenolid, a bioactive compound with cytostatic effect on cancer cells and deterrent effect on larvae, under conditions facilitating mating, which reflects a defect in chemical communication. Strains lacking either of the MAPkinases become female sterile, indicating the conservation of the role of MAPkinases in sexual fertility also in T. reesei. In summary, our findings substantiate the previously detected interconnection of cellulase regulation with regulation of secondary metabolism as well as the involvement of MAPkinases in light dependent gene regulation of cellulase and secondary metabolite genes in fungi., b

Summary:

Therefore, we asked whether the regulatory function of the MAPkinases might be connected to the role of VEL1 by testing transcript abundance of vel1 in deletion strains of tmk1, tmk2 and tmk3. Indeed, we found a light dependent regulation of vel1 in all MAPkinase mutants, with differential impacts either in constant light or in constant darkness. The previously detected only minor effect of TMK2 on cbh1 transcript abundance may be due to the uncontrolled light conditions during cultivation: Since we observed a clear increase of cbh1 in tmk2 in darkness and a decrease in light, what previously was found, may well be a mixture of these effects. In case of TMK3 our results for cellulase regulation are in agreement with previous data, although also here the regulation pattern we observed is more severe, with activity and transcript levels barely detectable anymore. We therefore checked available transcriptome data from comparable conditions for indications if such a feedback might exist,, but since we did not find significant regulation of tmk1, tmk2 or tmk3 in these data, we conclude that this is not the case. Interestingly, in N. crassa the OS pathway, corresponding to the Hog1-pathway in yeast and comprising a homologue of TMK3 has no significant influence on cellulase production, which is in contrast to our results. In summary, our data obtained with experiments under controlled light conditions clearly show a light dependent regulatory function of all three MAPkinases on cellulase gene regulation and secreted cellulase activity, which is jeopardized by random light pulses. The GPCR CSG1, which is essential for the chemotropic response of T. reesei to glucose, was shown to be required for posttranscriptional regulation of cellulase gene expression. All three mutants were confirmed to only have a single integration of the deletion cassette by copy number determination. All crosses for the analysis of sexual development were performed on 60 mm 2% MEX agar plates at 22 C and 12 h lightdark cycles as previously described. At least three biological replicates were considered in every assay. For NMR spectroscopic measurements -bisorbibutenolide was dissolved in CD3OD and transferred into 5 mm high precision NMR sample tubes.

Figures:

Figure 1: Relevance of MAPkinases for phenotype and biomass formation.

Figure 2: Relevance of MAPkinases for chemotropic response and cellulase regulation.

Figure 3: Relevance of MAPkinases for sorbicillin production and genes involved in secondary metabolism.

Figure 4: Involvement of MAPkinases in mating abilities.

Figure 5: HPLC analysis of MAPK deletion mutants and identification of sorbicillin derivatives.

Figure 6: Schematic representation of the involvement of the MAPKinases, TMK1, TMK2 and TMK3 in sexual development (female fertility), cellulase regulation and secondary metabolism in constant light (LL) and constant darkness (DD).

Mentioned biomedical entities:

Genes: CSG2, CWI, Fus3, Git1, Gpa2, MAPK, MAT1-1, MAT1-2, MPK1, Slt2, Tmk1, Tmk2, VEL1, XYR1, cbh1, cellulase, cre1, env1, peroxidase, pks4, sar1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. flavus

Tissues: Ascospore, capillary

Pathways: N/A

Query Result 10/25

Functional Characterization of Sugar Transporter CRT1 Reveals Differential Roles of Its C-Terminal Region in Sugar Transport and Cellulase Induction in Trichoderma reesei

Wang, Zhixing; Yang, Renfei; Lv, Wenhao; Zhang, Weixin; Meng, Xiangfeng; Liu, Weifeng; Atanasova, Lea

Microbiology Spectrum

Year: 2022 PMID: 35852347 doi: 10.1128/spectrum.00872-22 Citations: 7 Score: 15.83

Keywords hits: cellulase: 67, reesei: 18, cbh1: 6, xyr1: 45

Abstract:

The expression of cellulase genes in lignocellulose-degrading fungus Trichoderma reesei is induced by insoluble cellulose and its soluble derivatives. Membrane-localized transporter/transceptor proteins have been thought to be involved in nutrient uptake and/or sensing to initiate the subsequent signal transduction during cellulase gene induction. Crt1 is a sugar transporter proven to be essential for cellulase gene induction although the detailed mechanism of Crt1-triggered cellulase induction remains elusive. In this study, we focused on the C-terminus region of Crt1 which is predicted to exist as an unstructured cytoplasmic tail in T. reesei. Serial C-terminal truncation of Crt1 revealed that deleting the last half of the C-terminal region of Crt1 hardly affected its transporting activity or ability to mediate the induction of cellulase gene expression. In contrast, removal of the entire C-terminus region eliminated both activities. Of note, Crt1-C5, retaining only the first five amino acids of C-terminus, was found to be capable of transporting lactose but failed to restore cellulase gene induction in the crt1 strain. Analysis of the cellular localization of Crt1 showed that Crt1 existed both at the plasma membrane and at the periphery of the nucleus although the functional relevance is not clear at present. Finally, we showed that the cellulase production defect of crt1 was corrected by overexpressing Xyr1, indicating that Xyr1 is a potential regulatory target of the signaling cascade initiated from Crt1.

Summary:

Particularly, while Crt1-C with the removal of the entire C-terminus lost its ability to transport lactose, Crt1-C5 that retains only the first five amino acids of the C-terminus was still transport capable. To further investigate whether the necessity of Crt1 in mediating cellulase induction absolutely depends on its transporting activities, the competence of the constructed Crt1 mutants in inducing cellulase production was evaluated by complementing the crt1 strain with the wild-type and mutant Crt1 under the control of a copper responsive promoter Ptcu1. Remarkable upregulation in the transcription of the main cellulase genes and the key transcriptional activator xyr1 was detected at 8 h for Re Crt1, Re Crt1-C22, and Re Crt1-CKR strains, which is much earlier than the parental strain TU-6, indicating that constitutive expression of crt1 facilitated the transcription of cellulase genes on lactose. The induced transcription of cellulase genes was also detected in Re Crt1 strain under Avicel inducing conditions however, the timing of cellulase gene transcription in Re Crt1 was not as advanced as that on lactose (Fig. We found that the transcription of crt1 is induced by Avicel in a similar pattern with cellulase genes. The transcription of crt1 was further analyzed in this strain on glucose.

Figures:

Figure 1: Subcellular localization of Crt1-GFP under cellulase induction condition.

Figure 2: Lactose transport activity analysis of wild type Crt1 and its C-terminal mutants.

Figure 3: Functional complementation test of Crt1 C-terminal mutants in the crt1 strain.

Figure 4: Transcriptional analysis of major cellulase genes and xyr1 in TU-6, crt1 , and different complementary strains under lactose inducing condition.

Figure 5: The carbon catabolite repression on the cellulase gene transcription was not fully relieved by overexpression of crt1 .

Figure 6: Xyr1 is essential but not sufficient for the transcription of crt1 .

Figure 7: Overexpression of Xyr1 in crt1 bypassed its cellulase expression defect.

Mentioned biomedical entities:

Genes: ADH1, AscI, CDT-2, Crt1, EGFR, ER, GLUT1, HXT, LAC12, Mep2, Rgt2, SspI, Stp1, Xyr1, actin, cbh1, cbh2, cellulase, eg1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 11/25

Regulation of Cellulase and Hemicellulase Gene Expression in Fungi

Amore, Antonella; Giacobbe, Simona; Faraco, Vincenza

Current Genomics

Year: 2013 PMID: 24294104 doi: 10.2174/1389202911314040002 Citations: 0 Score: 15.27

Keywords hits: cellulase: 180, reesei: 50, cbh1: 15, xyr1: 30

Abstract:
Research on regulation of cellulases and hemicellulases gene expression may be very useful for increasing the production of these enzymes in their native producers. Mechanisms of gene regulation of cellulase and hemicellulase expression in filamentous fungi have been studied, mainly in Aspergillus and Trichoderma. The production of these extracellular enzymes is an energy-consuming process, so the enzymes are produced only under conditions in which the fungus needs to use plant polymers as an energy and carbon source. Moreover, production of many of these enzymes is coordinately regulated, and induced in the presence of the substrate polymers. In addition to induction by mono- and oligo-saccharides, genes encoding hydrolytic enzymes involved in plant cell wall deconstruction in filamentous fungi can be repressed during growth in the presence of easily metabolizable carbon sources, such as glucose. Carbon catabolite repression is an important mechanism to repress the production of plant cell wall degrading enzymes during growth on preferred carbon sources. This manuscript reviews the recent advancements in elucidation of molecular mechanisms responsible for regulation of expression of cellulase and hemicellulase genes in fungi.

Summary:
Consistent with its role as the major transcriptional regulator of cellulase gene expression, a strongly increased basal expression of xyr1 was observed in strain CL847, which was further induced by lactose. However, significant amounts of xylanase were also produced when cellulose powder was used as a carbon source. In 2012, Sun et al. showed that N. crassa responds to the presence of cellulose by inducing both cellulase and hemicellulase gene expression, while the exposure to xylan only induces hemicellulase gene expression. Two zinc binuclear cluster transcription factors are important regulators of genes encoding both cellulases and hemicellulases in the presence of cellulose as carbon source, but they are not required for growth or hemicellulase activity production in the presence of xylan as reported by Coradetti et al. . Although the result was preliminary, the authors suggested the existence of a link in the regulation of the carbon and nitrogen utilization pathways in filamentous fungi.3.2 Sun et al. investigated CCR of cellulase expression in N. crassa and they showed that, under cellulolytic conditions, CRE-1 regulates genes involved in plant cell wall utilization by directly binding to adjacent motifs in promoter regions and also may compete for binding with positive regulatory factors. In accordance with these findings, a recent study by Gremel et al. indeed revealed that cellulase gene expression can be enhanced by the addition of methionine. PacC is the major factor involved in pH-dependent expression in Aspergillus spp. pH regulation of genes encoding cell wall-degrading enzymes has not been studied in detail in Aspergillus.

Figures:

Mentioned biomedical entities:

Genes: ACE1, ACE2, Ace2, AceI, AreA, CBH, CCR, CEL3A, CEL5A, CEL6A, CEL7A, CLR-1, CPC1, CRE-1, CRE1, ClbR, CreA, CreB, CreD, GATA-1, HAP2, HAP3, HAP5, LAE1, LIM1, NF-YA, NFYB, NIT2, PacC, SNF1, TATA, XYR1, XlnR, Xyr1, ace1, ace2, areA, cbh1, cbh2, cel6a, cellulase, clbR, cmc1, cre1, cre2, gh10-2, gsn, hap2, histone, pelA, rglA, swo1, xlnD, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. aculeatus, A. niger, A. oryzae, A. phoenicis, A. terreus

Tissues: N/A

Pathways: N/A

Query Result 12/25

Ras GTPases Modulate Morphogenesis, Sporulation and Cellulase Gene Expression in the Cellulolytic Fungus Trichoderma reesei, bRas Modulate Morphogenesis and Cellulase Formation

Zhang, Jiwei; Zhang, Yanmei; Zhong, Yaohua; Qu, Yinbo; Wang, Tianhong; Xue, Chaoyang

PLoS ONE

Year: 2012 PMID: 23152805 doi: 10.1371/journal.pone.0048786 Citations: 0 Score: 15.19

Keywords hits: cellulase: 69, reesei: 50, cbh1: 25, xyr1: 47

Abstract:
The model cellulolytic fungus Trichoderma reesei (teleomorph Hypocrea jecorina) is capable of responding to environmental cues to compete for nutrients in its natural saprophytic habitat despite its genome encodes fewer degradative enzymes. Efficient signalling pathways in perception and interpretation of environmental signals are indispensable in this process. Ras GTPases represent a kind of critical signal proteins involved in signal transduction and regulation of gene expression. In T. reesei the genome contains two Ras subfamily small GTPases TrRas1 and TrRas2 homologous to Ras1 and Ras2 from S. cerevisiae, but their functions remain unknown.

Summary:
However, detailed functions of these two Ras signal proteins in this organism remain unknown. It is also known that in many fungi, e. g., S. cerevisiae, C. neoformans and C. albicans, Ras plays an important role in activating the cAMP pathway to regulate cell morphology and cell cycle. Taken together, these results suggest that Tr Ras1 is an important signal protein that controls hyphal cell formation and polarized apical growth in T. reesei deficiency of the hyphal growth subsequently results in a small and dense colony in the Tr Ras1 deletion mutant Tr Ras1 is involved in controlling asexual development in T. reesei. These data suggest that Tr Ras2 may sense the signals except cellulose and act upstream of the transcription regulators to modulate cellulase gene expression. It is believed that the expression of cellulolytic gene is regulated by the cellulase transcription factors in T. reesei, brid=pone.0048786-Stricker2 ref-type=bibr13,,,. Thus, it seems that Tr Ras2 signalling may transmit the light or other signals to modulate cellulase gene expression. It has been discovered that cAMP signalling is involved in regulation of cellulase gene expression,. However, no alteration in the cAMP level on cellulose or glucose is detected in the mutant with a deletion for Tr Ras2, indicating that influence of Tr Ras2 on cellulase gene transcription is independent of cAMP signalling pathway. Xyr1 has been demonstrated as a central transcriptional regulator that controls xylanolytic as well as cellulolytic enzyme genes expression in T. reesei, brid=pone.0048786-Stricker2 ref-type=bibr13.

Figures:

Figure 1: Phylogenetic analysis of Ras proteins.

Figure 2: Analysis of transcript levels ofTrRas1andTrRas2mRNA by qRT-PCR.

Figure 3: Growth of TU-6, TrRas1and TrRas2on different carbon sources.

Figure 4: TrRas1 regulates filamentation and sporulation inT. reesei.

Figure 5: TrRas2 modulates polarized apical growth, branch formation and sporulation inT. reesei.

Figure 6: TrRas2 influences cellulase formation inT. reesei.

Figure 7: Analysis of the influence of TrRas2 on expression of the transcription factors encoding genes.

Figure 8: Influence of Xyr1 overexpression in theTrRas2deletion strain on regulation of cellulase gene transcription.

Figure 9: TrRas1 and TrRas2 play similar roles in increasing cAMP content.

Figure 10: Schematic model depicting TrRas1/2 signalling in regulating morphogenesis and cellulase gene expression inT. reesei.

Mentioned biomedical entities:

Genes: Ace1, Ace2, Cdc42, Cellulase, Cre1, GNA3, MAP, MAPK, Ras, Ras1, Ras2, RasA, RasB, RasC, Rho, Xyr1, ace1, ace2, actin, cAMP, cbh1, cbh2, cellulase, cre1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, hyphal cell

Organisms: C. albicans, C. neoformans, human

Tissues: peripheral

Pathways: N/A

Query Result 13/25

Nucleo-cytoplasmic shuttling dynamics of the transcriptional regulators XYR1 and CRE1 under conditions of cellulase and xylanase gene expression in Trichoderma reesei

Lichius, Alexander; Seidl-Seiboth, Verena; Seiboth, Bernhard; Kubicek, Christian P

Molecular Microbiology

Year: 2014 PMID: 25302561 doi: 10.1111/mmi.12824 Citations: 0 Score: 15.00

Keywords hits: cellulase: 74, reesei: 29, cbh1: 9, xyr1: 163

Abstract:

Trichoderma reesei is a model for investigating the regulation of (hemi-)cellulase gene expression. Cellulases are formed adaptively, and the transcriptional activator XYR1 and the carbon catabolite repressor CRE1 are main regulators of their expression. We quantified the nucleo-cytoplasmic shuttling dynamics of GFP-fusion proteins of both transcription factors under cellulase and xylanase inducing conditions, and correlated their nuclear presence/absence with transcriptional changes. We also compared their subcellular localization in conidial germlings and mature hyphae. We show that cellulase gene expression requires de novo biosynthesis of XYR1 and its simultaneous nuclear import, whereas carbon catabolite repression is regulated through preformed CRE1 imported from the cytoplasmic pool. Termination of induction immediately stopped cellulase gene transcription and was accompanied by rapid nuclear degradation of XYR1. In contrast, nuclear CRE1 rapidly decreased upon glucose depletion, and became recycled into the cytoplasm. In mature hyphae, nuclei containing activated XYR1 were concentrated in the colony center, indicating that this is the main region of XYR1 synthesis and cellulase transcription. CRE1 was found to be evenly distributed throughout the entire mycelium. Taken together, our data revealed novel aspects of the dynamic shuttling and spatial bias of the major regulator of (hemi-)cellulase gene expression, XYR1, in T. reesei.

Summary:

Nuclear recruitment of XYR1 and nuclear loss of CRE1, respectively, were triggered by addition of sophorose to germlings pre-cultured for 22 h in glucose, whereas nuclear loss of XYR1 and nuclear recruitment of CRE1, respectively, were initiated by addition of glucose to germlings pre-cultured for 44 h on cellulose. Nuclear recruitment of XYR1 was detectable within 5 min after addition of sophorose, however, required about 30 min more until a significantly high increase in nuclear signal could be observed. The cytoplasmic pool of XYR1 did not increase during this period, suggesting either that significant export of GFPXYR1 from the nucleus did not occur, or that GFP, as stable end-product of GFPXYR1 degradation, remained inside the nucleus. Together these data suggest that, in contrast to XYR1, CRE1 is being recycled. It is thus possible that CRE1 initially has to be exported from the nucleus to initiate its degradation, while this proteolysis later on also continues in the nucleus or acts immediately on the protein transported into the cytoplasm. The Western blot data revealed that the degradation of XYR1 is actually much faster. Together with our observation that residual amounts of glucose do not interfere with the inductive carbon source, and thus do not terminate cellulase gene expression, this suggests a mechanism by which no terminating signal might be required to switch off cellulase and hemicellulase gene expression, but that solely depends on the presence of the inducer. The mechanism that leads to nuclear export and degradation of CRE1, therefore, remains to be determined. Our data also demonstrated that XYR1 is not equally distributed in the hyphal nuclei, and in fact occurs mainly in nuclei of the central area of the colony.

Figures:

Figure 1: Nuclear import of XYR1 was significantly increased under cellulase inducing conditions.A and B.

Figure 2: Protein turn-over kinetics of both fusion reporters differ, leading to significant nuclear accumulation of free GFP from GFPXYR1 degradation.A.

Figure 3: XYR1 and CRE1 showed distinct nuclear shuttling dynamics.

Figure 4: XYR1 nuclear import dynamics correlated with gene expression.A.

Figure 5: Complete inhibition of XYR1 nuclear import and cellulase gene expression by cycloheximide treatment.

Figure 6: Transcription factor recruitment in the functionally stratified colony.A.

Mentioned biomedical entities:

Genes: ACE1, ACE2, AflR, AmyR, CRE1, Cellulase, Cre1, CreA, Gal1, Gal4, Leu3, MA, Mig1, NirA, Oaf1, Put3, Snf1, War1, XYR1, XlnR, Xyr1, cbh1, cellulase, cre1, gpd1, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. niger, mouse

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 14/25

Analysis of Light- and Carbon-Specific Transcriptomes Implicates a Class of G-Protein-Coupled Receptors in Cellulose Sensing

Stappler, Eva; Dattenbck, Christoph; Tisch, Doris; Schmoll, Monika; Mitchell, Aaron P.

mSphere

Year: 2017 PMID: 28497120 doi: 10.1128/mSphere.00089-17 Citations: 28 Score: 14.50

Keywords hits: cellulase: 81, reesei: 28, cbh1: 9, xyr1: 3

Abstract:

In fungi, most metabolic processes are subject to regulation by light. For Trichoderma reesei, light-dependent regulation of cellulase gene expression is specifically shown. Therefore, we intended to unravel the relationship between regulation of enzymes by the carbon source and regulation of enzymes by light. Our two-dimensional analysis included inducing and repressing carbon sources which we used to compare light-specific regulation to dark-specific regulation and to rule out effects specific for a single carbon source. We found close connections with respect to gene regulation as well as significant differences in dealing with carbon in the environment in light and darkness. Moreover, our analyses showed an intricate regulation mechanism for substrate degradation potentially involving surface sensing and provide a basis for knowledge-based screening for strain improvement.

Summary:

We did this analysis separately in light and darkness and found that induction-specific regulation does occur in light and darkness but that the number of genes with induction-specific regulation only in darkness or only in light is considerable. Genes associated with repression of cellulase gene expression upon growth on glucose versus growth on glycerol along with their functional categories are outlined in the supplemental material. The BLR1, BLR2, and ENV1 photoreceptors as well as their homologues in N. crassa were shown to be involved in regulation of expression of cellulolytic enzymes. Only 218 genes were found to be induction specific in light and darkness and not regulated by the photoreceptors. Of the 383 genes upregulated under inducing conditions in light and darkness, 61 genes are positively regulated by BLR1 in light. This analysis showed that with respect to regulation of substrate degradation, photoreceptors clearly contribute to regulation of induction-specific genes, including enzyme-encoding genes, in light and darkness.

Figures:

Figure 1: Experimental approach and light-specific regulation.

Figure 2: Clustering of genome-wide regulation patterns.

Figure 3: Analysis of GPCR deletion strains.

Figure 4: Model of CSG1 function.

Mentioned biomedical entities:

Genes: BLR1, BLR2, CSG2, Cellulase, CreC, ENV1, GNA1, GNA3, GNG1, GPR39, LIM1, MAT1-1, WD40, ace2, axe1, bgl1, blr1, cbh1, cel12a, cel1b, cellulase, chi18-9, cip1, cip2, cre1, gut1, hfb2, trehalase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: humans

Tissues: photoreceptor

Pathways: N/A

Query Result 15/25

Novel genetic tools that enable highly pure protein production in Trichoderma reesei

Rantasalo, Anssi; Vitikainen, Marika; Paasikallio, Toni; Jntti, Jussi; Landowski, Christopher P.; Mojzita, Dominik

Scientific Reports

Year: 2019 PMID: 30902998 doi: 10.1038/s41598-019-41573-8 Citations: 27 Score: 14.38

Keywords hits: cellulase: 18, reesei: 50, cbh1: 27, xyr1: 2

Abstract:
Trichoderma reesei is an established protein production host with high natural capacity to secrete enzymes. The lack of efficient genome engineering approaches and absence of robust constitutive gene expression systems limits exploitation of this organism in some protein production applications. Here we report engineering of T. reesei for high-level production of highly enriched lipase B of Candida antarctica (calB) using glucose as a carbon source. Multiplexed CRISPR/Cas9 in combination with the use of our recently established synthetic expression system (SES) enabled accelerated construction of strains, which produced high amounts of highly pure calB. Using SES, calB production levels in cellulase-inducing medium were comparable to the levels obtained by using the commonly employed inducible cbh1 promoter, where a wide spectrum of native enzymes were co-produced. Due to highly constitutive expression provided by the SES, it was possible to carry out the production in cellulase-repressing glucose medium leading to around 4 grams per liter of fully functional calB and simultaneous elimination of unwanted background enzymes.

Summary:
The cellobiohydrolase I was not selected for this test, since the cbh1 locus is a common site for targeted integration of the heterologous gene expression cassette, which results in the cbh1 coding region replacement by the target gene, utilizing the cbh1 promoter for the expression in the final production strain. In a single transformation event, two gRNAs targeting the Cas9 complex into the 5- and 3- regions of each gene were used together with three donor DNA molecules, each encoding pyr4 marker gene flanked with DNA homologous to the genomic regions outside the Cas9 target sites. However, in case of the CBHI-calB strain, in SG-lactose medium, a more pronounced drop in the transcription was observed for the sTF in the fifth day of cultivation, which correlated with the lower expression of calB in this time point. Previously it was observed that the expression of genes involved in protein folding and secretion closely follows the expression pattern of the cellulolytic genes, being repressed in presence of glucose and induced in lactose. However, due to inefficient homologous recombination in T. reesei, performing the targeted genetic changes is a tedious and time consuming process. The use of CRISPR/Cas9 has been successfully established in T. reesei, where up to three simultaneous genes deletions were demonstrated in a strain carrying a Cas9 expression cassette integrated in the genome. Cultivation media samples for subsequent analysis were collected on days 1, 3 and 5. The second bioreactor cultivation set was done using the calB expressing SES strains to compare the functionality of a short signal sequence and the CBHI-secretion-carrier in the SG-lactose medium fed-batch cultures in 1 L bioreactors. The cultivations were conducted for 5 days following the same principles as described above, the cultivation media samples were collected on days 2 to 5, and mycelium samples for the transcription analysis were collected on days 3 and 5. The culture supernatant samples collected during the bioreactor cultivations were diluted into water prior to loading into precast 420% Criterion SDS-PAGE gels.

Figures:

Figure 1: Scheme of the CRISPR/Cas9-mediated multiplexed gene deletions in T .

Figure 2: Comparison of approaches for the calB production in T .

Figure 3: Production of highly enriched calB protein using a short secretion signal sequence.

Mentioned biomedical entities:

Genes: Cas9, ER, Kex2, XYR1, act1, calB, cbh1, cbh2, cellulase, lipase, pdi1, sTF, sec61, ubc, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: protoplast

Organisms: N/A

Tissues: tube

Pathways: N/A

Query Result 16/25

Global Reprogramming of Gene Transcription in Trichoderma reesei by Overexpressing an Artificial Transcription Factor for Improved Cellulase Production and Identification of Ypr1 as an Associated Regulator

Zhang, Fei; Li, Jia-Xiang; Champreda, Verawat; Liu, Chen-Guang; Bai, Feng-Wu; Zhao, Xin-Qing

Frontiers in Bioengineering and Biotechnology

Year: 2020 PMID: 32719779 doi: 10.3389/fbioe.2020.00649 Citations: 0 Score: 14.21

Keywords hits: cellulase: 117, reesei: 156, cbh1: 5, xyr1: 12

Abstract:

Synthetic biology studies on filamentous fungi are providing unprecedented opportunities for optimizing this important category of microbial cell factory. Artificial transcription factor can be designed and used to offer novel modes of regulation on gene transcription network. Trichoderma reesei is commonly used for cellulase production. In our previous studies, a plasmid library harboring genes encoding artificial zinc finger proteins (AZFPs) was constructed for engineering T. reesei, and the mutant strains with improved cellulase production were selected. However, the underlying mechanism by which AZFP function remain unclear. In this study, a T. reesei Rut-C30 mutant strain T. reesei U5 bearing an AZFP named as AZFP-U5 was focused, which secretes high level protein and shows significantly improved cellulase and xylanase production comparing with its parental strain. In addition, enhanced sugar release was achieved from lignocellulosic biomass using the crude cellulase from T. reesei U5. Comparative transcriptome analysis was further performed, which showed reprogramming of global gene transcription and elevated transcription of genes encoding glycoside hydrolases by overexpressing AZFP-U5. Furthermore, 15 candidate regulatory genes which showed remarkable higher transcription levels by AZFP-U5 insertion were overexpressed in T. reesei Rut-C30 to examine their effects on cellulase biosynthesis. Among these genes, TrC3093861 (ypr1) and TrC3074374 showed stimulating effects on filter paper activity (FPase), but deletion of these two genes did not affect cellulase activity. In addition, increased yellow pigment production in T. reesei Rut-C30 by overexpression of gene ypr1 was observed, and changes of cellulase gene transcription were revealed in the ypr1 deletion mutant, suggesting possible interaction between pigment production and cellulase gene transcription. The results in this study revealed novel aspects in regulation of cellulase gene expression by the artificial regulators. In addition, the candidate genes and processes identified in the transcriptome data can be further explored for synthetic biology design and metabolic engineering of T. reesei to enhance cellulase production.

Summary:

Meanwhile, transcription of the major transcription factor xyr1 gene also increased, of which the transcript in T. reesei U5 was 5-fold to that in Rut-C30 at 48 h. The transcription levels of other regulators for cellulase regulation were also remarkably increased, which include ace1, ace2, and bglr. The artificial regulator AZFP-U5 is composed of C2H2 type zinc finger and Gal4 effector domain. At 48 h, the transcription of extracellular major glycoside hydrolase genes in the two strains decreased significantly compared to that at 24 h, and the total FPKM values of extracellular protein U5 and Rut-C30 at 48 h decreased to 2225% of that at 24 h, meanwhile, the transcripts of genes encoding the major cellulases from GH5, GH6, and GH7 families occupied 11.2% of the total cellulase transcripts in cellulase Rut-C30, 33.6% in cellulase U5. Transcripts of genes in the ERAD pathway, as well as ubiquitination and proteasome-related genes increased remarkably, indicating that the protein processing of T. reesei U5 has begun speedily at 24 h, which resulted in a large increase in extracellular cellulase protein secretion in T. reesei U5 compared to that in T. reesei Rut-C30. In addition, there are several significantly different transcriptional levels in metabolic pathways or cellular physiological activities between T. reesei U5 and Rut-C30 at 24 h, and the transcript level of genes in the following groups always improved remarkably in T. reesei U5: RNA polymerase II subunit and its basic transcriptional regulatory factors, mRNA regulatory pathways, RNA degradation, peroxisomes, CoA synthesis pathways, N-glycoside synthesis pathways, ammonia Acyl-tRNA synthesis pathway, nicotinamide metabolism pathway, fatty acid metabolism pathway, endoplasmic reticulum membrane protein processing, and proteasome complex. In the transcriptome data analysis of T. reesei U5, putative regulatory genes changing significantly at both 24 and 48 h are likely to be targets for activating the transcription of cellulase genes. Meanwhile, the transcripts of genes encoding the major cellulase regulatory factors, such as the genes of Xyr1 and Ace3, have increased significantly, except for Vib1. In our previous work, T. reesei strains overexpressing AZFP-U3 and AZFP-M2 showed enhanced cellulase production. Therefore, it is of great interest to further explore the mechanisms by which this AZFP functions. In the previous studies, transcriptome analysis was performed using T. reesei grown in different carbon source and lignocellulose substrates, and data analysis has been focused on the functions of known major regulators, such as Xyr1 and Cre1.

Figures:

Figure 1: Comparison of FPase activities and secreted protein concentration between the mutant T. reesei U5 and the parental strain Rut-C30.

Figure 2: Hydrolysis of pretreated corn stalk by cellulase produced by T. reesei U5 and Rut-C30.

Figure 3: Transcription of genes encoding major cellulases and transcription factors in T. reesei U5 and Rut-C30.

Figure 4: Gene Ontology analysis for DEGs between T. reesei U5 and Rut-C30.

Figure 5: Functional categories for DEGs between T. reesei U5 and Rut-C30.

Figure 6: Differential transcription of genes encoding potential transcription regulators (A) and CAZymes (B) .

Figure 7: Effects of overexpression of 15 transcription factors on cellulase production Cellulase (A) , xylanase and protein productions (B) by T. reesei Rut-C30 recombinant strains overexpressing the 15 candidate transcription factors.

Figure 8: Effects of ypr1 on pigment production and sporulation.

Figure 9: Proposed model for the regulatory effect of AZF-U5 on cellulase synthesis.

Mentioned biomedical entities:
Genes: APCS, Ace2, Ace3, CBH, CBH1, Cdt1, Cdt2, Cellulase, Clr2, CltB, Cre1, Crt1, Ctf1, Gal4, HSP70, LysR, MA, U5, UBE1C, Vel2, Vib1, Xyr1, Ypr1, ace1, ace2, axe1, bgl1, cbh1, cbh2, cellulase, cip1, crt1, eg1, hac1, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: pulp

Pathways: N/A

Query Result 17/25

Enhancement of cellulase production in Trichoderma reesei RUT-C30 by comparative genomic screening

Liu, Pei; Lin, Aibo; Zhang, Guoxiu; Zhang, Jiajia; Chen, Yumeng; Shen, Tao; Zhao, Jian; Wei, Dongzhi; Wang, Wei

Microbial Cell Factories

Year: 2019 PMID: 31077201 doi: 10.1186/s12934-019-1131-z Citations: 16 Score: 13.47

Keywords hits: cellulase: 82, reesei: 52, cbh1: 2, xyr1: 5

Abstract:
Cellulolytic enzymes produced by the filamentous fungus Trichoderma reesei are commonly used in biomass conversion. The high cost of cellulase is still a significant challenge to commercial biofuel production. Improving cellulase production in T. reesei for application in the cellulosic biorefinery setting is an urgent priority.

Summary:
Genome sequencing revealed that unlike SS-II, which displayed higher cellulase production compared with that of RUT-C30, as measured by FPase, SS-II possesses wild-type cre1. Mutation of cre1 is not always the key step to increase total cellulase production in T. reesei because of the effect on the growth of T. reesei. The 57 genes with specific mutations in SS-II might be involved in cellulase hyper-production of SS-II. Of these five mutants, the 108642 strain displayed an 83.7% increase in FPase activity, which was the highest extracellular cellulase activity observed among the tested strains. The activity was approximately 13.26/nmol/min/mg using NADP+ as the coenzyme. One unit of enzymatic activity was defined as 1 mol of p-nitrophenol released from the substrate per min. The concentration of proteins in the supernatant was measured using a commercial protein assay kit based on the absorbance at 595 nm. Hydrolysis of pretreated corn stover was performed as previously described .

Figures:

Figure 1: Cellulase production of SS-II- cre1 96 .

Figure 2: Cellulase activity and secreted protein concentration of T. reesei mutants.

Figure 3: Expression levels of cellulase related genes in T. reesei mutants.

Mentioned biomedical entities:

Genes: CRE1, Cellulase, GRD1, MA, PCS, bgl1, cbh1, cbh2, cellulase, cre1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 18/25

Genome sequencing of the Trichoderma reesei QM9136 mutant identifies a truncation of the transcriptional regulator XYR1 as the cause for its cellulase-negative phenotype

Lichius, Alexander; Bidard, Frdrique; Buchholz, Franziska; Le Crom, Stphane; Martin, Joel; Schackwitz, Wendy; Austerlitz, Tina; Grigoriev, Igor V; Baker, Scott E; Margeot, Antoine; Seiboth, Bernhard; Kubicek, Christian P

BMC Genomics

Year: 2015 PMID: 25909478 doi: 10.1186/s12864-015-1526-0 Citations: 0 Score: 13.24

Keywords hits: cellulase: 49, reesei: 42, cbh1: 2, xyr1: 210

Abstract:
Trichoderma reesei is the main industrial source of cellulases and hemicellulases required for the hydrolysis of biomass to simple sugars, which can then be used in the production of biofuels and biorefineries. The highly productive strains in use today were generated by classical mutagenesis. As byproducts of this procedure, mutants were generated that turned out to be unable to produce cellulases. In order to identify the mutations responsible for this inability, we sequenced the genome of one of these strains, QM9136, and compared it to that of its progenitor T. reesei QM6a.

Summary:
As shown in Figure A and B, transformation with the gfp-xyr1 fusion gene fully restored the typical growth defective phenotype of XYR1 loss-of-function mutants on 65 mM D-xylose and matched the average colony extension speed as well as macroscopical morphology of T. reesei QM9414 and QM9414 transformed with the identical GFP-XYR1 construct. To test whether nuclear import of GFP-XYR1 in a QM9136 background is also sufficient to restore cellulase and hemicellulase gene expression, we performed correlated gene expression analysis on mycelia from liquid cultures induced by 1.4 mM sophorose. This finding suggests that the truncated XYR1 protein undergoes proteasomal degradation much faster than the full length version, probably because it is quickly detected as non-functional. Despite the strong reduction of the degree and rate of nuclear uptake in the strain harbouring the xyr1A2294 allele, the cytosolic level of the truncated XYR1 remained low but comparable to that in the parental strain. This finding is also nicely reflected in the phenotype of QM9136, which fails to grow on 65 mM D-xylose, has strongly reduced or no growth on lactose and cellulose, respectively, , and is deficient in the formation of cellulases, xylanases and mannanases . While the identification of a gene that is already known to be crucially involved in cellulase and hemicellulase formation, as the cause for the inability of cellulase formation of T. reesei QM9136, may be considered trivial, it nevertheless sheds new light into its molecular function: XYR1 belongs to the fungal-specific Zn2 Cys6-binuclear zinc cluster transcription factor family ,. A mutation in or absence of this coiled-coil domain therefore may impair binding of XYR1, and thus cellulase and hemicellulase gene expression. Interestingly, two mutations have been reported in the N-terminal part of the acidic activation domain that result in the opposite, i. e. constitutive expression of xylanases in A. niger and T. reesei and elevated basal expression of cellulases in T. reesei only. Growth of T. reesei QM6a and QM9136 on solid minimal agar medium supplemented with either D-glucose, cellobiose or lactose. Additional file 2: Table S1. Chromosomal location of mutated genes in QM9136. Additional file 3: Figure S2. cDNA alignment of native and mutated xyr1 loci.

Figures:

Figure 1: Gene replacement with full-length GFP-XYR1 construct rescues XYR1-loss-of-function phenotype of QM9136.

Figure 2: Gene replacement with the truncated GFP-XYR1 1780 reproduced the XYR1-loss-of-function phenotype in QM9414.

Figure 3: Nuclear import of GFP-XYR1 is indistinguishable in QM9136 and QM9414 transformants.

Figure 4: Western blot analysis of GFP-XYR1 and GFP-XYR1 1780 under cellulase inducing conditions.

Figure 5: C-terminal truncation renders GFP-XYR1 1780 non-responsive to inductive signals.

Figure 6: Quantification of nuclear recruitment of GFP-XYR1 and GFP-XYR1 1780 in the three main functional zones of the mature colony.

Mentioned biomedical entities:

Genes: ACE2, ACE3, B, CRE1, ENS, HD, MA, SEB, XYR1, XlnR, Xyr1, cbh1, cellulase, cre1, ptrA, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. niger, mouse

Tissues: lens, tube

Pathways: N/A

Query Result 19/25

The -importin KAP8 (Pse1/Kap121) is required for nuclear import of the cellulase transcriptional regulator XYR1, asexual sporulation and stress resistance in Trichoderma reesei

Ghassemi, Sara; Lichius, Alexander; Bidard, Frderique; Lemoine, Sophie; Rossignol, Marie-Nolle; Herold, Silvia; Seidl-Seiboth, Verena; Seiboth, Bernhard; Espeso, Eduardo A; Margeot, Antoine; Kubicek, Christian P

Molecular Microbiology

Year: 2015 PMID: 25626518 doi: 10.1111/mmi.12944 Citations: 0 Score: 13.23

Keywords hits: cellulase: 45, reesei: 87, cbh1: 4, xyr1: 55

Abstract:

The ascomycete Trichoderma reesei is an industrial producer of cellulolytic and hemicellulolytic enzymes, and serves as a prime model for their genetic regulation. Most of its (hemi-)cellulolytic enzymes are obligatorily dependent on the transcriptional activator XYR1. Here, we investigated the nucleo-cytoplasmic shuttling mechanism that transports XYR1 across the nuclear pore complex. We identified 14 karyopherins in T.reesei, of which eight were predicted to be involved in nuclear import, and produced single gene-deletion mutants of all. We found KAP8, an ortholog of Aspergillus nidulansKapI, and Saccharomyces cerevisiaeKap121/Pse1, to be essential for nuclear recruitment of GFP-XYR1 and cellulase gene expression. Transformation with the native gene rescued this effect. Transcriptomic analyses of kap8 revealed that under cellulase-inducing conditions 42 CAZymes, including all cellulases and hemicellulases known to be under XYR1 control, were significantly down-regulated. kap8 strains were capable of forming fertile fruiting bodies but exhibited strongly reduced conidiation both in light and darkness, and showed enhanced sensitivity towards abiotic stress, including high osmotic pressure, low pH and high temperature. Together, these data underscore the significance of nuclear import of XYR1 in cellulase and hemicellulase gene regulation in T.reesei, and identify KAP8 as the major karyopherin required for this process.

Summary:

Using a systematic gene deletion approach, we identified KAP8, the ortholog of Pse1/Kap121from S. cerevisiae and A. nidulans KapI, respectively, to be required for XYR1 import into the nucleus, and demonstrate that this step is essential for cellulase and hemicellulase gene expression. B, expression of cbh1 in the kap8 strain was only 2% of that shown by the complemented transformant strain. We therefore concluded that kap8 is essential for cellulase gene expression and while it cannot be ruled out that other importins can also transport XYR1 into the nucleus is of major importance for the function of XYR1. In order to test whether KAP8 is indeed essential for XYR1 uptake into the nucleus, we expressed a GFP-XYR1 fusion protein in T. reesei kap8 and its complemented transformant and monitored its subcellular localization under inducing conditions. In T. reesei, sporulation of the kap8 strain was also reduced to 10% of that of the parental strain (n = 4 Fig. Table lists all T. reesei strains used and produced in this study. The A. nidulans strains used in this study are given in Supporting Information Table.

Figures:

Figure 1: Expression of the T .

Figure 2: Nuclear recruitment of GFP-XYR1 was not observed in the kap8 strain; however, it was restored in the kap8 complemented transformat (kap8 ct).

Figure 3: Number of genes, grouped according to functional similarity, that were induced by sophorose at least fourfold (grey bars) and genes whose sophorose induction was significantly lower (5-fold) in the kap8 strain compared with the complemented transformant kap8ct (white bars).

Figure 4: Growth of A .

Figure 5: Effect of kap8 knock out on asexual and sexual development in T .

Figure 6: Growth of T .

Figure 7: Inventory of gene groups that were not induced by sophorose but significantly down-regulated (at least 5-fold; P 0.05) in the kap8 strain compared with the retransformant.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, Aft1, AlcR, B, CRE1, Cellulase, Gal4, KAP1, KAP3, KAP6, KAP8, Kap104, MAT1-1, MAT1-2, NirA, Nmd5, Pdr1, Pho4, Pse1, Ran, Sln1, Ste12, Sxm1, XYR1, XlnR, Yap1, cbh1, cel7A, cellulase, kap1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 20/25

Role of cellulose response transporter-like protein CRT2 in cellulase induction in Trichoderma reesei

Yan, Su; Xu, Yan; Yu, Xiao-Wei

Biotechnology for Biofuels and Bioproducts

Year: 2023 PMID: 37488642 doi: 10.1186/s13068-023-02371-7 Citations: 0 Score: 13.20

Keywords hits: cellulase: 117, reesei: 36, cbh1: 2, xyr1: 43

Abstract:
Induction of cellulase in cellulolytic fungi Trichoderma reesei is strongly activated by cellulosic carbon sources. The transport of cellulosic inducer and the perception of inducing signal is generally considered as the critical process for cellulase induction, that the inducing signal would be perceived by a sugar transporter/transceptor in T. reesei. Several sugar transporters are coexpressed during the induction stage, but which function they serve and how they work collaboratively are still difficult to elucidate.

Summary:
D), which might be the reason for the comparative cellulase production from 72 h. Consistent with the cellulase activity in 77517, the transcription of cel7a in 77517 was similar to the level in the parent strain, which further confirmed that the deletion of tre77517 does not affect cellulase induction on Avicel. C), and the transcription of the major regulator xyr1 and transporter crt1 were also upregulated when mycelia were exposed to cellobiose and lactose, similar to the phenotype in Avicel medium. Besides, the expression of crt2 on Avicel shows a similar trend to the expression of crt1, and xyr1-overexpression could activate the expression of crt2 and crt1 in the early stage of cellulase induction. Among these sugar transporters, the cellobiose/lactose transporter CRT1 is critical for cellulase induction in T. reesei, and the disruption of CRT1 impaired cellulase induction either on cellulose or soluble lactose . In this study, we characterized a novel cellulose response transporter-like protein CRT2, which is not capable for sugar transport, but shares slight similarity to CRT1 which transduce inducing signal through activating the major cellobiose transporter CRT1 and the transcription factor XYR1. Previous study conducted by Porciuncula et al. indicated a delayed lactose uptake when crt2 was disrupted , but our study proved that CRT2 cannot transport lactose and cellobiose (Fig.

Figures:
Figure 1: Evaluating the effect on the dysfunction of sugar transporters for cellulase production.

Figure 2: Analysis of sugar transport of TRE77517 in S. cerevisiae .

Figure 3: Subcellular distribution of TRE77517GFP in T. reesei .

Figure 4: Sugar transporter-like protein TRE77517 regulates cellulase induction.

Figure 5: Dysfunction of CRT2 (TRE77517) did not affect cellobiose/lactose consumption.

Figure 6: CRT2 functions in a CRT1-like manner in cellulase induction by activating CRT1 and XYR1.

Figure 7: Activated cellulase induction through overexpressed crt2 was still repressed by CCR.

Figure 8: Diagram of the cellulase induction by CRT2, CRT1 and XYR1 in T. reesei .

Mentioned biomedical entities:
Genes: CCR, CDT-2, CLP1, CRE1, CRT1, Cellulase, Crt1, GLT-1, GLT1, SNF3, STP1, URA3, XYR1, cdt-2, cellulase, clp1, cre1, crt1, gh1-1, sar1, vps13, vps21

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: E. coli, Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 21/25

Gene Co-expression Network Reveals Potential New Genes Related to Sugarcane Bagasse Degradation in Trichoderma reesei RUT-30

Borin, Gustavo Pagotto; Carazzolle, Marcelo Falsarella; dos Santos, Renato Augusto Corra; Riao-Pachn, Diego Mauricio; Oliveira, Juliana Velasco de Castro

Frontiers in Bioengineering and Biotechnology

Year: 2018 PMID: 30406095 doi: 10.3389/fbioe.2018.00151 Citations: 0 Score: 13.00

Keywords hits: cellulase: 28, reesei: 52, cbh1: 9, xyr1: 44

Abstract:

The biomass-degrading fungus Trichoderma reesei has been considered a model for cellulose degradation, and it is the primary source of the industrial enzymatic cocktails used in second-generation (2G) ethanol production. However, although various studies and advances have been conducted to understand the cellulolytic system and the transcriptional regulation of T. reesei, the whole set of genes related to lignocellulose degradation has not been completely elucidated. In this study, we inferred a weighted gene co-expression network analysis based on the transcriptome dataset of the T. reesei RUT-C30 strain aiming to identify new target genes involved in sugarcane bagasse breakdown. In total, 70% of all the differentially expressed genes were found in 28 highly connected gene modules. Several cellulases, sugar transporters, and hypothetical proteins coding genes upregulated in bagasse were grouped into the same modules. Among them, a single module contained the most representative core of cellulolytic enzymes (cellobiohydrolase, endoglucanase, -glucosidase, and lytic polysaccharide monooxygenase). In addition, functional analysis using Gene Ontology (GO) revealed various classes of hydrolytic activity, cellulase activity, carbohydrate binding and cation:sugar symporter activity enriched in these modules. Several modules also showed GO enrichment for transcription factor activity, indicating the presence of transcriptional regulators along with the genes involved in cellulose breakdown and sugar transport as well as other genes encoding proteins with unknown functions. Highly connected genes (hubs) were also identified within each module, such as predicted transcription factors and genes encoding hypothetical proteins. In addition, various hubs contained at least one DNA binding site for the master activator Xyr1 according to our in silico analysis. The prediction of Xyr1 binding sites and the co-expression with genes encoding carbohydrate active enzymes and sugar transporters suggest a putative role of these hubs in bagasse cell wall deconstruction. Our results demonstrate a vast range of new promising targets that merit additional studies to improve the cellulolytic potential of T. reesei strains and to decrease the production costs of 2G ethanol.

Summary:

Since then, RUT-C30 strain has been the target of numerous studies on the conversion of biomass into biofuels and other high-value products. Previously, Borin et al. investigated the transcriptome of T. reesei RUT-C30 grown on steam-exploded sugarcane bagasse in a time course of 6, 12, and 24 h as well as on fructose after 24 h. Interestingly, a set of cellulase, hemicellulase, TF, and sugar transporter coding genes were activated in bagasse along with genes encoding hypothetical or uncharacterized proteins. In this study, an extensive number of genes were found to be co-expressed in bagasse, and they could be the target of new studies to evaluate their role in lignocellulose degradation. Gene co-expression network analysis was performed based on a RNA-Seq dataset from a previous study conducted by our research group. In addition, 14 putative sugar transporters, three Zn2 Cys6 transcriptional regulators, one putative GCN5-acetyltransferase, one methyltransferase and several other genes were also co-expressed with xyr1 and had XBS in their promoters. Finally, hub genes that were neighbor nodes to the master activator Xyr1 and that had XBS in their promoter were also identified. As a result, 28 modules of co-expressed genes were formed, and only eight were chosen for further analysis due to the largest number of up and downregulated genes encoding CAZymes, TFs, sugar transporters and other genes of unknown function that were co-expressed. After the MCODE clustering, the GO enrichment showed that several terms related to lignocellulose breakdown, including polysaccharide and cellulose binding, and hydrolase activities, were enriched in the subcluster 1 of the coral1 module where the master activator Xyr1 was also found. In addition to the cbh1 and xyn1, the promoter of the cbh2 gene also had two XBS predicted in this study that were shared with its homologs in other T. reesei strains. Based on the assumption that Xyr1 is essential to the induction of the CAZymes and sugar transporters, it was not surprising that genes related to carbohydrate transport and metabolism, such as CAZymes and putative sugar transporters, were the most abundant upregulated genes that had XBS in the KOG analyses.

Figures:

Figure 1: DEGs identified in the modules of the T. reesei RUT-C30 network encoding CAZyme, sugar transporter, transcription factor and unknown proteins with signal peptide.

Figure 2: Putative XBS predicted in the promoter of 22 genes encoding cellulases and hemicellulases and chosen to be sought in the gene promoter regions of T. reesei .

Figure 3: KOG function classification of the DEGs identified in the up and down sets that had at least one predicted XBS.

Figure 4: (A) Upregulated genes encoding proteins of different functions being co-expressed with the activator xyr1 in the coral1 module.

Mentioned biomedical entities:

Genes: Ama1, AreA, CBM1, Cip1, Cre1, ER, GPI, Hsf1, MFS, Pac1, PacC, Pro1, Ras, Swo1, TRAP, WetA, Xyn1, Xyr1, bgl1, bgl2, cbh1, cellulase, cip1, cre1, env1, histone, pac1, pro1, swo1, wetA, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: spore

Pathways: N/A

Query Result 22/25

Induction of cellulase production by Sr2+ in Trichoderma reesei via calcium signaling transduction

Li, Ni; Zeng, Yi; Chen, Yumeng; Shen, Yaling; Wang, Wei

Bioresources and Bioprocessing

Year: 2022 PMID: 38647894 doi: 10.1186/s40643-022-00587-3 Citations: 0 Score: 12.41

Keywords hits: cellulase: 84, reesei: 29, cbh1: 10, xyr1: 10

Abstract:

Trichoderma reesei RUT-C30 is a well-known high-yielding cellulase-producing fungal strain that converts lignocellulose into cellulosic sugar for resource regeneration. Calcium is a ubiquitous secondary messenger that regulates growth and cellulase production in T. reesei. We serendipitously found that adding Sr2+ to the medium significantly increased cellulase activity in the T. reesei RUT-C30 strain and upregulated the expression of cellulase-related genes. Further studies showed that Sr2+ supplementation increased the cytosolic calcium concentration and activated the calcium-responsive signal transduction pathway of Ca2+calcineurin-responsive zinc finger transcription factor 1 (CRZ1). Using the plasma membrane Ca2+ channel blocker, LaCl3, we demonstrated that Sr2+ induces cellulase production via the calcium signaling pathway. Supplementation with the corresponding concentrations of Sr2+ also inhibited colony growth. Sr2+ supplementation led to an increase in intracellular reactive oxygen species (ROS) and upregulated the transcriptional levels of intracellular superoxide dismutase (sod1) and catalase (cat1). We further demonstrated that ROS content was detrimental to cellulase production, which was alleviated by the ROS scavenger N-acetyl cysteine (NAC). This study demonstrated for the first time that Sr2+ supplementation stimulates cellulase production and upregulates cellulase genes via the calcium signaling transduction pathway. Sr2+ leads to an increase in intracellular ROS, which is detrimental to cellulase production and can be alleviated by the ROS scavenger NAC. Our results provide insights into the mechanistic study of cellulase synthesis and the discovery of novel inducers of cellulase.

Summary:

These results indicate that supplementation with large amounts of Sr2+ inhibited the growth of T. reesei strains under both glucose and Avicel carbon sources. To evaluate the effect of Sr2+ on cellulase production, the same amount of RUT-C30 mycelia was transferred into liquid MM with 1% Avicel as the carbon source and different concentrations of Sr2+. The optimal concentration of Sr2+ for enhancing cellulase production was 70 mM in RUT-C30, which was used in subsequent analyses. To further investigate the effect of Sr2+ on cellulase synthesis, the expression levels of two vital cellulase genes and essential cellulase transcription activators xyr1 and ace3 were detected using RT-qPCR, as shown in Fig. After addition of 70 mM Sr2+, the expression of sod1 increased by approximately 78150% at 4860 h. The transcript levels of cat1 increased by 165487% at 6072 h, indicating that the strain was under high ROS pressure. NAC and H2O2 were used to alter the intracellular ROS content. According to previous studies, Ca2+ is an essential secondary messenger that can cooperate with intracellular cAMP, highosmolarity glycerol and ROS to regulate life activities in filamentous fungi, which is worthy of further study. In addition, large amounts of ROS were produced after treatment with 70 mM Sr2+ based on analysis using an ROS probe.

Figures:

Figure 1: Effect of different concentrations of Sr 2+ on hyphal growth in T. reesei RUT-C30 strain.

Figure 2: Effect of Sr 2+ on cellulase production in RUT-C30 strain.

Figure 3: Effect of Sr 2+ on cellulase-related gene transcription levels in RUT-C30 strain.

Figure 4: Cytosolic Ca 2+ levels increase after Sr 2+ addition.

Figure 5: Effect of LaCl 3 on cellulase production after Sr 2+ treatment.

Figure 6: Increase in cytosolic ROS level after Sr 2+ addition.

Figure 7: Effect of ROS on Sr 2+ -induced cellulase production.

Figure 8: Mechanistic model of Sr 2+ -induced cellulase overexpression in T. reesei .

Mentioned biomedical entities:

Genes: CaM, CnA, Crz1, XYR1, cbh1, cellulase, sar1, sod1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Ganoderma lucidum

Tissues: N/A

Pathways: N/A

Query Result 23/25

The transcription factor STE12 influences growth on several carbon sources and production of dehydroacetic acid (DHAA) in Trichoderma reesei

Schalamun, Miriam; Hinterdobler, Wolfgang; Schinnerl, Johann; Brecker, Lothar; Schmoll, Monika

Scientific Reports

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Keywords hits: cellulase: 16, reesei: 37, cbh1: 13, xyr1: 6

Abstract:

The filamentous ascomycete Trichoderma reesei, known for its prolific cellulolytic enzyme production, recently also gained attention for its secondary metabolite synthesis. Both processes are intricately influenced by environmental factors like carbon source availability and light exposure. Here, we explore the role of the transcription factor STE12 in regulating metabolic pathways in T. reesei in terms of gene regulation, carbon source utilization and biosynthesis of secondary metabolites. We show that STE12 is involved in regulating cellulase gene expression and growth on carbon sources associated with iron homeostasis. STE12 impacts gene regulation in a light dependent manner on cellulose with modulation of several CAZyme encoding genes as well as genes involved in secondary metabolism. STE12 selectively influences the biosynthesis of the sorbicillinoid trichodimerol, while not affecting the biosynthesis of bisorbibutenolide, which was recently shown to be regulated by the MAPkinase pathway upstream of STE12 in the signaling cascade. We further report on the biosynthesis of dehydroacetic acid (DHAA) in T. reesei, a compound known for its antimicrobial properties, which is subject to regulation by STE12. We conclude, that STE12 exerts functions beyond development and hence contributes to balance the energy distribution between substrate consumption, reproduction and defense.

Summary:

Strong connections were observed for light response and regulation of plant cell wall degradation, but also secondary metabolism is influenced by light and carbon sources as is sexual development,. STE12 and STE12-like transcription factors are unique to fungi and well-known as targets of the mating/pheromone MAPkinase pathway,. AE). We investigated the regulatory impact of STE12 on gene regulation upon growth on cellulose as carbon source in constant light and constant darkness.).1D and 2D NMR measurements of the 7% dehydroacetic acid in sample A further confirmed the structure of the target compound. Therefore, a contribution of STE12 to transmission of the signal regulating pks4 by the cell integrity pathway in light would not be without precedent. Our analysis of carbon source utilization, i. e. growth on diverse carbon sources in darkness, did not reveal dramatic alterations in growth of ste12, indicating that STE12 is not essential for the considerable adaptation competence of the metabolism of T. reesei. The specific script developed for and used in this analysis is available at: https://github. com/miriamschalamun/RNATricho/tree/main. Statistical significance for RTqPCR, cellulase activity and biomass analysis was calculated in R using Students T-test = p-value 0.01, = p-value 0.05. Variations in growth based on diverse carbon sources were assessed using the BIOLOG FF Microplate assay, as described previously.

Figures:

Figure 1: Characteristics of ste12 /TR36543/TrA1391C and its encoded protein.

Figure 2: Relevance of STE12 for biomass formation, cellulase activity and gene regulation upon growth on 1% cellulose.

Figure 3: Analysis of carbon source utilization using the BIOLOG phenotype microassay.

Figure 4: Gene ontology analysis of genes regulated by STE12.

Figure 5: HPLC analysis of secondary metabolite production and identification of dehydroacetic acid.

Figure 6: Schematic representation of the light dependent and overlapping targets of the MAPKinase pathway and STE12.

Mentioned biomedical entities:

Genes: Fet5, Fus3, Kss1, STE12, Ste12, VEL1, bgl1, cbh1, cellulase, pks2, pks4, sar1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: C. neoformans

Tissues: capillary, tube

Pathways: N/A

Query Result 24/25

Engineering the Effector Domain of the Artificial Transcription Factor to Improve Cellulase Production by Trichoderma reesei

Meng, Qing-Shan; Zhang, Fei; Wang, Wei; Liu, Chen-Guang; Zhao, Xin-Qing; Bai, Feng-Wu

Frontiers in Bioengineering and Biotechnology

Year: 2020 PMID: 32671045 doi: 10.3389/fbioe.2020.00675 Citations: 0 Score: 11.93

Keywords hits: cellulase: 68, reesei: 89, cbh1: 1, xyr1: 37

Abstract:

Filamentous fungal strains of Trichoderma reesei have been widely used for cellulase production, and great effort has been devoted to enhancing their cellulase titers for the economic biorefinery of lignocellulosic biomass. In our previous studies, artificial zinc finger proteins (AZFPs) with the Gal4 effector domain were used to enhance cellulase biosynthesis in T. reesei, and it is of great interest to modify the AZFPs to further improve cellulase production. In this study, the endogenous activation domain from the transcription activator Xyr1 was used to replace the activation domain of Gal4 of the AZFP to explore impact on cellulase production. The cellulase producer T. reesei TU-6 was used as a host strain, and the engineered strains containing the Xyr1 and the Gal4 activation domains were named as T. reesei QS2 and T. reesei QS1, respectively. Compared to T. reesei QS1, activities of filter paper and endoglucanases in crude cellulase produced by T. reesei QS2 increased 24.6 and 50.4%, respectively. Real-time qPCR analysis also revealed significant up-regulation of major genes encoding cellulase in T. reesei QS2. Furthermore, the biomass hydrolytic performance of the cellulase was evaluated, and 83.8 and 97.9% more glucose was released during the hydrolysis of pretreated corn stover using crude enzyme produced by T. reesei QS2, when compared to the hydrolysis with cellulase produced by T. reesei QS1 and the parent strain T. reesei TU-6. As a result, we proved that the effector domain in the AZFPs can be optimized to construct more effective artificial transcription factors for engineering T. reesei to improve its cellulase production.

Summary:

Therefore, it is of great interest to investigate the effect of endogenous effector domains in the AZFPs on regulation of cellulase production in T. reesei. In T. reesei, Xyr1 is the Gal4 family transcription activator that is essential for cellulase/hemicellulase gene transcription. Among these genes, the transcription of cel61a and cip2 in T. reesei QS2 were 4.1- and 2.7-fold as well as 19.7- and 2.3-fold higher than that from T. reesei QS1 and TU-6 at 24 h, respectively, suggesting that the crude enzyme secreted by T. reesei QS2 is more suitable for further hydrolysis of lignocellulosic biomass. In addition to the enzyme-encoding genes mentioned above, the transcriptional levels of the most known cellulase transcription factor genes were also significantly altered in both T. reesei QS1 and QS2. Among these transcription factor genes, xyr1, which encodes the major transcriptional activator for cellulase biosynthesis, was significantly up-regulated by 3.8- and 2.9- fold in T. reesei QS2 at 24 compared with that of T. reesei QS1 and the parent strain, respectively. It will be important to explore regulation of accessory protein biosynthesis by the AZFP developed in this study, so that further synthetic biology design can be performed to increase hydrolysis efficiency of lignocellulosic biomass. In this study, we designed a new artificial transcription factor AZFPM2-Xyr1AD in T. reesei, and found that AZFPM2-Xyr1AD could significantly improve the expression of major cellulase genes. As a result, we proved that the endogenous effector domain Xyr1AD is more advantageous for construction of AZFP not only in improving cellulase production, but also in optimizing cellulase complex for biomass conversion. All datasets generated for this study are included in the article/. X-QZ and Q-SM conceived the project and designed the experiments.

Figures:

Figure 1: Construction of fusion transcription factor AZFP M2 -Xyr1 AD in T. reesei .

Figure 2: Cellulase production by T. reesei QS2, QS1 and TU-6.

Figure 3: Transcription analysis of genes encoding major cellulolytic enzymes and regulators.

Figure 4: Hydrolysis of alkali pretreated corn stover (APCS) and jerusalem artichoke stalk (APJAS) by cellulases produced by T. reesei QS2, QS1, and TU-6.

Mentioned biomedical entities:

Genes: AD, ADs, APCS, Ace1, Ace2, Ace3, BglR, Cel61A, Cip1, Cip2, Cre1, Ctf1, Gal4, Rce1, SAGA, Swo1, Vib1, Xyr1, bgl1, cbh1, cbh2, cellulase, cip2, tef1, xyn3

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 25/25

Early cellular events and potential regulators of cellulase induction in Penicillium janthinellum NCIM 1366

Christopher, Meera; Sreeja-Raju, AthiraRaj; Abraham, Amith; Gokhale, Digambar Vitthal; Pandey, Ashok; Sukumaran, Rajeev K.

Scientific Reports

Year: 2023 PMID: 36977777 doi: 10.1038/s41598-023-32340-x Citations: 0 Score: 11.63

Keywords hits: cellulase: 109, reesei: 14, cbh1: 5, xyr1: 16

Abstract:
Cellulase production by fungi is tightly regulated in response to environmental cues, and understanding this mechanism is a key pre-requisite in the efforts to improve cellulase secretion. Based on UniProt descriptions of secreted Carbohydrate Active enZymes (CAZymes), 13 proteins of the cellulase hyper-producer Penicillium janthinellum NCIM 1366 (PJ-1366) were annotated as cellulases- 4 cellobiohydrolases (CBH), 7 endoglucanases (EG) and 2 beta glucosidases (BGL). Cellulase, xylanase, BGL and peroxidase activities were higher for cultures grown on a combination of cellulose and wheat bran, while EG was stimulated by disaccharides. Docking studies indicated that the most abundant BGL- Bgl2- has different binding sites for the substrate cellobiose and the product glucose, which helps to alleviate feedback inhibition, probably accounting for the low level of glucose tolerance exhibited. Out of the 758 transcription factors (TFs) differentially expressed on cellulose induction, 13 TFs were identified whose binding site frequencies on the promoter regions of the cellulases positively correlated with their abundance in the secretome. Further, correlation analysis of the transcriptional response of these regulators and TF-binding sites on their promoters indicated that cellulase expression is possibly preceded by up-regulation of 12 TFs and down-regulation of 16 TFs, which cumulatively regulate transcription, translation, nutrient metabolism and stress response.

Summary:
Since information on cellulase regulation in this genus is limited, we have analysed both the transcriptional and secretome response to induction, and have correlated this data with the distribution of transcription factor binding sites on the promoters of its cellulases so as to derive an understanding of the regulators and pathways that govern the early induction of cellulases. Fourteen substrates were studied for their effect on enzyme production- the monomers arabinose, glucose and xylose the disaccharides cellobiose, lactose, maltose and mannose lignin and its components gluconic acid and vanillin the polysaccharides cellulose and xylan wheat bran as a representative of a lignocellulosic substrate, and a combination of wheat bran and cellulose in the ratio 2.5:1. Of these, protein sequence data was available for 16 factors. On the whole, it was seen that in addition to known cellulase regulators like Cre1, Xyr1 and the Ace proteins, expression of cellulases also seems to be regulated by factors participating in transcriptional control, growth and cell cycle regulation, control of phosphate metabolism, and global regulators like Gcn4 and Ste12. The promoters of these 13 TFs, in turn, contained binding sites for a further 45 transcription factors. Since the abundance of BGL proteins in the CW secretome does not seem to correlate with the observed enzyme activity, it leads to 2 possibilities- the observed high pNPGase activity in the CW medium might be due to the action of other glycanases, or the two BGLs that are secreted in the CW medium have exceptionally high catalytic efficiency. In addition to identifying the secreted cellulases of PJ-1366 and the substrates suitable for their expression, this study has also attempted to identify the early regulators of cellulase expression, based on the presence of binding sites of known TFs on the cellulase promoters, and correlating their distribution with the transcriptome and secretome profiles of the fungus. The major features of the complex cellulase regulatory network has been studied in many species of Trichoderma, Neurospora and Aspergillus, in which the master regulator of cellulase expression is the positive regulator Xyr1 or its orthologs.

Figures:

Figure 1: Secreted protein and enzyme activity profiles of PJ-1366 grown on different carbon sources.

Figure 2: Protein abundance at 3, 7 and 13days, of PJ-1366 cultures grown on different substrates.

Figure 3: Activity of PJ-1366 BGLs in the presence of glucose.

Figure 4: Correlations in protein abundance vs distribution of TF-binding sites in their respective promoters.

Figure 5: Fold change in expression of proteins predicted as early regulators of cellulase gene expression.

Mentioned biomedical entities:

Genes: Abf1, Ace1, Ace2, Adr1, AreA, BGLs, Bgl1, Bgl2, BglX, CCR, CDT-2, CRT1, CW, Cat8, Cbf1, Cbh1, Cbh2, Cellulase, Clr1, Cpc1, Cre1, Ctr1, Ctr3, ER, Eg1, Eg2, FacB, Fbf1, GH, Gal4, Gcn4, Gcr1, Lac9, MLP, Mac1, Mcm1, Med8, Mig1, Nit2, Pdr1, Pdr3, Peroxidase, Pho4, Rap1, Rcs1, Reb1, Rox1, Rpn4, SAGA, Sip4, Sko1, Ste12, Sut1, Tbp, Ume6, Xbp1, XylE, Xyr1, ace2, bgl1, cbh1, cellulase, cre1, peroxidase, rRNA, xylE

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A